

## 2-4-5

1. Let  $f$  be the function defined by  $f(x) = k\sqrt{x} - \ln x$ , for  $x > 0$ , where  $k$  is a positive constant. Find  $f'(x)$  and  $f''(x)$ .

## Part A

1 point is earned for:  $f'(x)$

1 point is earned for:  $f''(x)$

$$f'(x) = \frac{k}{2\sqrt{x}} = -\frac{1}{x}$$

$$f''(x) = -\frac{1}{4}kx^{-3/2} + x^{-2}$$



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## 2-4-5

2. Let  $f$  be a function that is differentiable throughout its domain and that has the following properties.

(i)  $f(x + y) = \frac{f(x) + f(y)}{1 - f(x)f(y)}$  for all real numbers  $x$ ,  $y$ , and  $x + y$  in the domain of  $f$

(ii)  $\lim_{h \rightarrow 0} f(h) = 0$

(iii)  $\lim_{h \rightarrow 0} \frac{f(h)}{h} = 1$

(a) Show that  $f(0) = 0$ .

(b) Use the definition of the derivative to show that  $f'(x) = 1 + [f(x)]^2$ . Indicate clearly where properties (i), (ii) and (iii) are used.

(c) Find  $f(x)$  by solving the differential equation in part (b).

## Part A

1 point is earned for to show  $f(0) = 0$

Method 1:

$f$  is continuous at 0, so  $f(0) = \lim_{x \rightarrow 0} f(x) = 0$

Method 2:

$$f(0) = f(0 + 0) = \frac{f(0) + f(0)}{1 - f(0)f(0)}$$

$$f(0) (1 - [f(0)]^2) = 2f(0)$$

$$f(0) (-1 - [f(0)]^2) = 0$$

$$f(0) = 0$$



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Method 2:

## 2-4-5

$$f(0) = f(0 + 0) = \frac{f(0) + f(0)}{1 - f(0)f(0)}$$

$$f(0) \left(1 - [f(0)]^2\right) = 2f(0)$$

$$f(0) \left(-1 - [f(0)]^2\right) = 0$$

$$f(0) = 0$$

**Part B**

1 point is earned for definition of  $f'(x)$

1 point is earned for clearly uses (i)

1 point is earned for clearly uses (ii)

1 point is earned for clearly uses (iii)

1 point is earned for algebraic manipulation to answer

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{\frac{f(x) + f(h)}{1 - f(x)f(h)} - f(x)}{h} \quad [\text{By (i)}] \\ &= \lim_{h \rightarrow 0} \left[ \frac{f(h)}{h} \cdot \frac{1 + [f(x)]^2}{1 - f(x)f(h)} \right] \\ &= 1 \cdot \frac{1 + [f(x)]^2}{1 - f(x) \cdot 0} \quad [\text{By (iii) \& (ii)}] \\ &= 1 + [f(x)]^2 \end{aligned}$$



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1 point is earned for clearly uses (i)

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1 point is earned for algebraic manipulation to answer

## 2-4-5

$$\begin{aligned}
 f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\
 &= \lim_{h \rightarrow 0} \frac{\frac{f(x)+f(h)}{1-f(x)f(h)} - f(x)}{h} && [\text{By (i)}] \\
 &= \lim_{h \rightarrow 0} \left[ \frac{f(h)}{h} \cdot \frac{1+[f(x)]^2}{1-f(x)f(h)} \right] \\
 &= 1 \cdot \frac{1+[f(x)]^2}{1-f(x) \cdot 0} && [\text{By (iii) \& (ii)}] \\
 &= 1 + [f(x)]^2
 \end{aligned}$$

**Part C**

1 point is earned if separates variables

1 point is earned for integration, with  $C$

1 point is earned for  $c$  evaluated

Method 1:

$$\text{Let } y = f(x); \frac{dy}{dx} = 1 + y^2$$

$$\frac{dy}{1+y^2} = dx$$

$$\tan^{-1} y = x + C$$

$$y = \tan(x + C)$$

$$f(0) = 0 \Rightarrow C = 0 \text{ [or } C = n\pi, n \in \mathbb{Z}]$$

$$f(x) = \tan x \text{ [or } f(x) = \tan(x + n\pi)]$$

or

Method 2:

Guess that  $f(x) = \tan x$

$$1 + [f(x)]^2 = 1 + \tan^2 x = \sec^2 x = f'(x)$$

$$f(0) = \tan(0) = 0$$

Since the solution to the D.E. is unique  $f(x) = \tan x$  is the solution.



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**2-4-5**

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Method 2:

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Since the solution to the D.E. is unique  $f(x) = \tan x$  is the solution.

## 2-4-5

3. Consider the curve defined by  $x^2 + xy + y^2 = 27$ .

- (a) Write an expression for the slope of the curve at any point  $(x, y)$ .
- (b) Determine whether the lines tangent to the curve at the  $x$ -intercepts of the curve are parallel. Show the analysis that leads to your conclusion.
- (c) Find the points on the curve where the lines tangent to the curve are vertical.

**Part A**

2 points are earned for the correct implicit differentiation

<-1> each incorrect derivative of a term

$$2x + xy' + y + 2yy' = 0$$

1 point is earned for correctly solving for  $y'$

0/1 if only one  $y'$

$$y' = \frac{-2x - y}{x + 2y}$$



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0/1 if only one  $y'$

$$y' = \frac{-2x - y}{x + 2y}$$

**Part B**

1 point is earned for correctly using  $y = 0$

**2-4-5**

If  $y = 0, x^2 = 27$

$x = \pm 3\sqrt{3}$

1 point is earned for correctly finding  $y'$  at  $x$ -intercepts

at  $x = 3\sqrt{3}, y' = \frac{-2 \cdot 3\sqrt{3}}{3\sqrt{3}} = -2$

at  $x = -3\sqrt{3}, y' = \frac{2 \cdot 3\sqrt{3}}{-3\sqrt{3}} = -2$

1 point is earned for the correct analysis and conclusion

Tangent lines at  $x$ -intercepts are parallel

NOTE: For solution at  $y$ -intercepts, eligible for 2nd and 3rd points



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1 point is earned for the correct analysis and conclusion

Tangent lines at  $x$ -intercepts are parallel

NOTE: For solution at  $y$ -intercepts, eligible for 2nd and 3rd points

**Part C**

1 point is earned for correctly setting  $x + 2y = 0$

$y'$  undefined if  $x + 2y = 0$

**2-4-5**

1 point is earned for correctly solving for  $y$  (or  $x$ ) values

$$\begin{aligned}(-2y)^2 + (-2y)y + y^2 &= 27 \\ 3y^2 &= 27 \\ y &= \pm 3\end{aligned}$$

1 point is earned for correctly giving both points

Points are  $(-6, 3)$  and  $(6, -3)$

NOTE: max 1/3 for horizontal tangents

Max 1/3 if denominator of  $y'$  is not of form  $ax + by + c$  with  $a \cdot b \neq 0$ .



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## 2-4-5

4. Consider the curve defined by  $x^2 + xy + y^2 = 27$ .  
Find the points on the curve where the lines tangent to the curve are vertical.

## Part C

1 point is earned for correctly setting  $x + 2y = 0$

$y'$  undefined if  $x + 2y = 0$

1 point is earned for correctly solving for  $y$  (or  $x$ ) values

$$(-2y)^2 + (-2y)y + y^2 = 27$$

$$3y^2 = 27$$

$$y = \pm 3$$

1 point is earned for correctly giving both points

Points are  $(-6, 3)$  and  $(6, -3)$

Note: max 1/3 for horizontal tangents

Max 1/3 if denominator of  $y'$  is not of form  $ax + by + c$  with  $a, b \neq 0$



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The student response earns three of the following points:

1 point is earned for correctly setting  $x + 2y = 0$

$y'$  undefined if  $x + 2y = 0$

1 point is earned for correctly solving for  $y$  (or  $x$ ) values

$$(-2y)^2 + (-2y)y + y^2 = 27$$

$$3y^2 = 27$$

$$y = \pm 3$$

1 point is earned for correctly giving both points

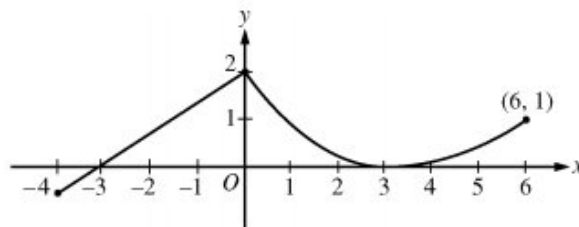
Points are  $(-6, 3)$  and  $(6, -3)$

Note: max 1/3 for horizontal tangents

Max 1/3 if denominator of  $y'$  is not of form  $ax + by + c$  with  $a, b \neq 0$

2-4-5

5.

Graph of  $f$ 

A continuous function  $f$  is defined on the closed interval  $-4 \leq x \leq 6$ . The graph of  $f$  consists of a line segment and a curve that is tangent to the  $x$ -axis at  $x = 3$ , as shown in the figure above. On the interval  $0 < x < 6$ , the function  $f$  is twice differentiable, with  $f''(x) > 0$ .

- (a) Is  $f$  differentiable at  $x = 0$ ? Use the definition of the derivative with one-sided limits to justify your answer.
- (b) For how many values of  $a$ ,  $-4 \leq a < 6$ , is the average rate of change of  $f$  on the interval  $[a, 6]$  equal to 0? Give a reason for your answer.
- (c) Is there a value of  $a$ ,  $-4 \leq a < 6$ , for which the Mean Value Theorem, applied to the interval  $[a, 6]$ , guarantees a value  $c$ ,  $a < c < 6$ , at which  $f'(c) = \frac{1}{3}$ ? Justify your answer.
- (d) The function  $g$  is defined by  $g(x) = \int_0^x f(t) dt$  for  $-4 \leq x \leq 6$ . On what intervals contained in  $[-4, 6]$  is the graph of  $g$  concave up? Explain your reasoning.

**Part A**

1 point is earned if sets up difference quotient at  $x = 0$

1 point is earned for the answer with justification

$$\lim_{h \rightarrow 0^-} \frac{f(h) - f(0)}{h} = \frac{2}{3}$$

$$\lim_{h \rightarrow 0^+} \frac{f(h) - f(0)}{h} < 0$$

Since the one-sided limits do not agree,  $f$  is not differentiable at  $x = 0$ .



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## 2-4-5

$$\lim_{h \rightarrow 0^-} \frac{f(h) - f(0)}{h} = \frac{2}{3}$$

$$\lim_{h \rightarrow 0^+} \frac{f(h) - f(0)}{h} < 0$$

Since the one-sided limits do not agree,  $f$  is not differentiable at  $x = 0$ .

**Part B**

1 point is earned for expression of average rate of change

1 point is earned for answer with reason

$$\frac{f(6) - f(a)}{6 - a} = 0 \text{ when } f(a) = f(6). \text{ There are two values of } a \text{ for which this is true.}$$



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The student response earns all of the following points:

1 point is earned for expression of average rate of change

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$$\frac{f(6) - f(a)}{6 - a} = 0 \text{ when } f(a) = f(6). \text{ There are two values of } a \text{ for which this is true.}$$

**Part C**

1 point is earned for answering "yes" and identifying  $a = 3$

1 point is earned for the justification

Yes,  $a = 3$ . The function  $f$  is differentiable on the interval  $3 < x < 6$  and continuous on  $3 \leq x \leq 6$ .

$$\text{Also, } \frac{f(6) - f(3)}{6 - 3} = \frac{1 - 0}{6 - 3} = \frac{1}{3}.$$

By the Mean Value Theorem, there is a value  $c$ ,  $3 < c < 6$ , such that  $f'(c) = \frac{1}{3}$ .



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The student response earns all of the following points:

1 point is earned for answering "yes" and identifying  $a = 3$

1 point is earned for the justification

**2-4-5**

Yes,  $a = 3$ . The function  $f$  is differentiable on the interval  $3 < x < 6$  and continuous on  $3 \leq x \leq 6$ .

Also,  $\frac{f(6)-f(3)}{6-3} = \frac{1-0}{6-3} = \frac{1}{3}$ .

By the Mean Value Theorem, there is a value  $c$ ,  $3 < c < 6$ , such that  $f'(c) = \frac{1}{3}$ .

**Part D**

1 point is earned for  $g'(x) = f(x)$

1 point is earned for considering  $g''(x) > 0$

1 point is earned for the answer

$$g'(x) = f(x), g''(x) = f'(x)$$

$$g''(x) > 0 \text{ when } f'(x) > 0$$

This is true for  $-4 < x < 0$  and  $3 < x < 6$ .



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| 0 | 1 | 2 | 3 |
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1 point is earned for considering  $g''(x) > 0$

1 point is earned for the answer

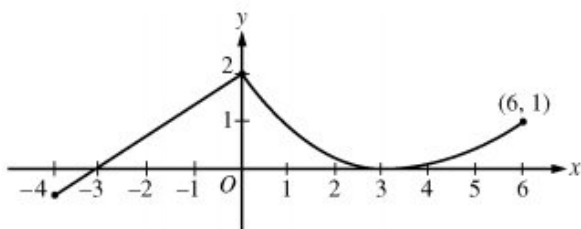
$$g'(x) = f(x), g''(x) = f'(x)$$

$$g''(x) > 0 \text{ when } f'(x) > 0$$

This is true for  $-4 < x < 0$  and  $3 < x < 6$ .

## 2-4-5

6.

Graph of  $f$ 

A continuous function  $f$  is defined on the closed interval  $-4 \leq x \leq 6$ . The graph of  $f$  consists of a line segment and a curve that is tangent to the  $x$ -axis at  $x=3$ , as shown in the figure above. On the interval  $0 < x < 6$ , the function  $f$  is twice differentiable, with  $f''(x) > 0$ .

Is  $f$  differentiable at  $x=0$ ? Use the definition of the derivative with one-sided limits to justify your answer.

**Part A**

1 point is earned if sets up difference quotient at  $x=0$

1 point is earned for the answer with justification

$$\lim_{h \rightarrow 0^-} \frac{f(h) - f(0)}{h} = \frac{2}{3}$$

$$\lim_{h \rightarrow 0^+} \frac{f(h) - f(0)}{h} < 0$$

Since the one-sided limits do not agree,  $f$  is not differentiable at  $x=0$ .



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The student response earns all of the following points:

1 point is earned if sets up difference quotient at  $x=0$

1 point is earned for the answer with justification

$$\lim_{h \rightarrow 0^-} \frac{f(h) - f(0)}{h} = \frac{2}{3}$$

$$\lim_{h \rightarrow 0^+} \frac{f(h) - f(0)}{h} < 0$$

Since the one-sided limits do not agree,  $f$  is not differentiable at  $x=0$ .

## 2-4-5

7. Let  $f$  be the function given by  $f(x)=|x|$ . Which of the following statements about  $f$  are true?

I.  $f$  is continuous at  $x=0$ .

II.  $f$  is differentiable at  $x=0$ .

III.  $f$  has an absolute minimum at  $x=0$ .

(A) I only

(B) II only

(C) III only

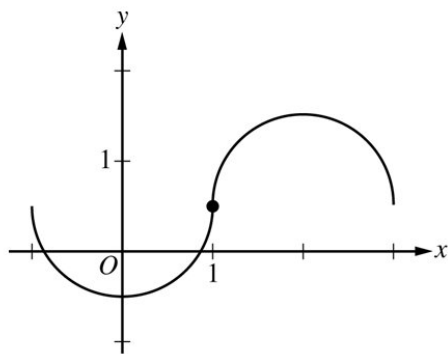
(D) I and III only

(E) II and III only



2-4-5

8.

Graph of  $h'$ 

The function  $h$  is defined on the closed interval  $[-1, 3]$ . The graph of  $h'$ , the derivative of  $h$ , is shown above. The graph consists of two semicircles with a common endpoint at  $x = 1$ . Which of the following statements about  $h$  must be true?

i.  $h(-1) = h(3)$

ii.  $h$  is continuous at  $x = 1$ .

iii. The graph of  $h$  has a vertical asymptote at  $x = 1$ .

(A) None

(B) II only

(C) I and II only

(D) I and III only



### Answer B

**This option is correct.** By the Fundamental Theorem of Calculus,  $h(3) = h(-1) + \int_{-1}^3 h'(x) dx$ . Since the region between the graph of  $h'$  and the  $x$ -axis has more area above the axis than below the axis, the definite integral  $\int_{-1}^3 h'(x) dx$  is positive. Therefore  $h(3) > h(-1)$  and statement I is false.

The graph shows that derivative of  $h$  exists on the interval  $(-1, 3)$ . That means that the function  $h$  is differentiable on  $(-1, 3)$  and is therefore continuous on that interval. In particular,  $h$  is continuous at  $x = 1$ , so statement II must be true. Because  $h$  is continuous at  $x = 1$  and  $h'(1) \gg 0.5$ , the graph of  $h$  has a tangent line at  $x = 1$  that is not vertical. Locally near  $x = 1$  the graph of  $h$  will look like that tangent line and thus there cannot be a vertical asymptote at  $x = 1$ . Therefore statement III is false.

## 2-4-5

9. Let  $f$  be the function defined by

$$f(x) = \begin{cases} \sqrt{x+1} & \text{for } 0 \leq x \leq 3 \\ 5-x & \text{for } 3 < x \leq 5. \end{cases}$$

- (a) Is  $f$  continuous at  $x = 3$ ? Explain why or why not.  
 (b) Find the average value of  $f(x)$  on the closed interval  $0 \leq x \leq 5$ .  
 (c) Suppose the function  $g$  is defined by

$$g(x) = \begin{cases} k\sqrt{x+1} & \text{for } 0 \leq x \leq 3 \\ mx+2 & \text{for } 3 < x \leq 5, \end{cases}$$

where  $k$  and  $m$  are constants. If  $g$  is differentiable at  $x = 3$ , what are the values of  $k$  and  $m$ ?

**Part A**

1 point is earned for answering "yes" and equating the values of the left- and right-hand limits

1 point is earned for the correct explanation involving limits

$f$  is continuous at  $x = 3$  because  $\lim_{x \rightarrow 3^-} f(x) = \lim_{x \rightarrow 3^+} f(x) = 2$ .

Therefore,  $\lim_{x \rightarrow 3} f(x) = 2 = f(3)$ .



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The student response earns all of the following points:

1 point is earned for answering "yes" and equating the values of the left- and right-hand limits

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Therefore,  $\lim_{x \rightarrow 3} f(x) = 2 = f(3)$ .

**Part B**

1 point is earned for  $k \int_0^3 f(x)dx + k \int_3^5 f(x)dx$  (where  $k \neq 0$ )

$$\int_0^5 f(x)dx = \int_0^3 f(x)dx + \int_3^5 f(x)dx$$

1 point is earned for the correct antiderivative of  $\sqrt{x+1}$

1 point is earned for the correct antiderivative of  $5 - x$

## 2-4-5

$$= \frac{2}{3}(x+1)^{3/2} \Big|_0^3 + \left(5x - \frac{1}{2}x^2\right) \Big|_3^5$$

$$= \left(\frac{16}{3} - \frac{2}{3}\right) + \left(\frac{25}{2} - \frac{21}{2}\right) = \frac{20}{3}$$

1 point is earned for the correct evaluation and answer

Average value:  $\frac{1}{5} \int_0^5 f(x) dx = \frac{4}{3}$



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The student response earns all of the following points:

1 point is earned for  $k \int_0^3 f(x) dx + k \int_3^5 f(x) dx$  (where  $k \neq 0$ )

$$\int_0^5 f(x) dx = \int_0^3 f(x) dx + \int_3^5 f(x) dx$$

1 point is earned for the correct antiderivative of  $\sqrt{x+1}$

1 point is earned for the correct antiderivative of  $5 - x$

$$= \frac{2}{3}(x+1)^{3/2} \Big|_0^3 + \left(5x - \frac{1}{2}x^2\right) \Big|_3^5$$

$$= \left(\frac{16}{3} - \frac{2}{3}\right) + \left(\frac{25}{2} - \frac{21}{2}\right) = \frac{20}{3}$$

1 point is earned for the correct evaluation and answer

Average value:  $\frac{1}{5} \int_0^5 f(x) dx = \frac{4}{3}$

**Part C**

1 point is earned for the correct answer  $2k = 3m + 2$

Since  $g$  is continuous at  $x = 3$ ,  $2k = 3m + 2$ .

$$g'(x) = \begin{cases} \frac{k}{2\sqrt{x+1}} & \text{for } 0 < x < 3 \\ m & \text{for } 3 < x < 5 \end{cases}$$

## 2-4-5

$$\lim_{x \rightarrow 3^-} g'(x) = \frac{k}{4} \text{ and } \lim_{x \rightarrow 3^+} g'(x) = m$$

1 point is earned for the correct answer  $\frac{k}{4} = m$

Since these two limits exist and  $g$  is differentiable at  $x = 3$ , the two limits are equal. Thus  $\frac{k}{4} = m$ .

1 point is earned for the correct values for  $k$  and  $m$

$$8m = 3m + 2; m = \frac{2}{5} \text{ and } k = \frac{8}{5}$$



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The student response earns all of the following points:

1 point is earned for the correct answer  $2k = 3m + 2$

Since  $g$  is continuous at  $x = 3$ ,  $2k = 3m + 2$ .

$$g'(x) = \begin{cases} \frac{k}{2\sqrt{x+1}} & \text{for } 0 < x < 3 \\ m & \text{for } 3 < x < 5 \end{cases}$$

$$\lim_{x \rightarrow 3^-} g'(x) = \frac{k}{4} \text{ and } \lim_{x \rightarrow 3^+} g'(x) = m$$

1 point is earned for the correct answer  $\frac{k}{4} = m$

Since these two limits exist and  $g$  is differentiable at  $x = 3$ , the two limits are equal. Thus  $\frac{k}{4} = m$ .

1 point is earned for the correct values for  $k$  and  $m$

$$8m = 3m + 2; m = \frac{2}{5} \text{ and } k = \frac{8}{5}$$

## 2-4-5

10. Let  $f$  be the function defined by  $f(x) = \begin{cases} \sqrt{x+1} & \text{for } 0 \leq x \leq 3 \\ 5-x & \text{for } 3 < x \leq 5. \end{cases}$

Suppose the function  $g$  is defined by

$$g(x) = \begin{cases} k\sqrt{x+1} & \text{for } 0 \leq x \leq 3 \\ mx+2 & \text{for } 3 < x \leq 5, \end{cases}$$

where  $k$  and  $m$  are constants. If  $g$  is differentiable at  $x = 3$ , what are the values of  $k$  and  $m$ ?

## Part C

1 point is earned for the correct answer  $2k = 3m + 2$

Since  $g$  is continuous at  $x = 3$ ,  $2k = 3m + 2$ .

$$g'(x) = \begin{cases} \frac{k}{2\sqrt{x+1}} & \text{for } 0 < x < 3 \\ m & \text{for } 3 < x < 5 \end{cases}$$

$$\lim_{x \rightarrow 3^-} g'(x) = \frac{k}{4} \text{ and } \lim_{x \rightarrow 3^+} g'(x) = m$$

1 point is earned for the correct answer  $\frac{k}{4} = m$

Since these two limits exist and  $g$  is differentiable at  $x = 3$ , the two limits are equal. Thus  $\frac{k}{4} = m$

1 point is earned for the correct values for  $k$  and  $m$

$$8m = 3m + 2; m = \frac{2}{5} \text{ and } k = \frac{8}{5}$$



|   |   |   |   |
|---|---|---|---|
| 0 | 1 | 2 | 3 |
|---|---|---|---|

The student response earns three of the following points:

1 point is earned for the correct answer  $2k = 3m + 2$

Since  $g$  is continuous at  $x = 3$ ,  $2k = 3m + 2$ .

$$g'(x) = \begin{cases} \frac{k}{2\sqrt{x+1}} & \text{for } 0 < x < 3 \\ m & \text{for } 3 < x < 5 \end{cases}$$

$$\lim_{x \rightarrow 3^-} g'(x) = \frac{k}{4} \text{ and } \lim_{x \rightarrow 3^+} g'(x) = m$$

1 point is earned for the correct answer  $\frac{k}{4} = m$

Since these two limits exist and  $g$  is differentiable at  $x = 3$ , the two limits are equal. Thus  $\frac{k}{4} = m$

1 point is earned for the correct values for  $k$  and  $m$

**2-4-5**

$$8m = 3m + 2; m = \frac{2}{5} \text{ and } k = \frac{8}{5}$$

## 2-4-5

11. Let  $f$  be a function defined by  $f(x) = \begin{cases} 2x - x^2 & \text{for } x \leq 1, \\ x^2 + kx + p & \text{for } x > 1. \end{cases}$
- (a) For what values of  $k$  and  $p$  will  $f$  be continuous and differentiable at  $x = 1$ ?
- (b) For the values of  $k$  and  $p$  found in part (a), on what interval or intervals is  $f$  increasing?
- (c) Using the values of  $k$  and  $p$  found in part (a), find all points of inflection of the graph of  $f$ . Support your conclusion.

**Part A**

1 point is earned for equation of  $f(1)$

1 point is earned for  $1 = 1 + k + p$

1 point is earned for  $0 = 2 + k$

1 point is earned for solution of system

For continuity at  $x = 1$ ,

$$\lim_{x \rightarrow 1^-} (2x - x^2) = f(1) = \lim_{x \rightarrow 1^+} (x^2 + kx + p)$$

Therefore  $1 = 1 + k + p$

Since  $f$  is continuous at  $x = 1$  and is piecewise polynomial, left and right derivatives exist.

$$f'_-(1) = 0 \text{ and } f'_+(1) = 2 + k$$

For differentiability at  $x = 1$ ,  $0 = 2 + k$ .

Therefore  $k = -2$ ,  $p = 2$ .



|   |   |   |   |   |
|---|---|---|---|---|
| 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|

The student response earns all of the following points:

1 point is earned for equation of  $f(1)$

1 point is earned for  $1 = 1 + k + p$

1 point is earned for  $0 = 2 + k$

1 point is earned for solution of system

For continuity at  $x = 1$ ,

## 2-4-5

$$\lim_{x \rightarrow 1^-} (2x - x^2) = f(1) = \lim_{x \rightarrow 1^+} (x^2 + kx + p)$$

Therefore  $1 = 1 + k + p$

Since  $f$  is continuous at  $x = 1$  and is piecewise polynomial, left and right derivatives exist.

$$f'_-(1) = 0 \text{ and } f'_+(1) = 2 + k$$

For differentiability at  $x = 1$ ,  $0 = 2 + k$ .

Therefore  $k = -2$ ,  $p = 2$ .

**Part B**

1 point is earned for correct  $f'$  and analysis of sign of  $f'$  on domain

1 point is earned for  $(-\infty, \infty)$

$$f'(x) = 2 - 2x, x \leq 1$$

$$2 - 2x > 0$$

$$x < 1$$

$$f'(x) = 2x - 2, x > 1$$

$$2x - 2 > 0$$

$$x > 1$$

Since  $f$  increases on each of  $(-\infty, 1)$  and  $(1, \infty)$  and is continuous at  $x = 1$ ,  $f$  is increasing on  $(-\infty, \infty)$ .



|   |   |   |
|---|---|---|
| 0 | 1 | 2 |
|---|---|---|

The student response earns all of the following points:

1 point is earned for correct  $f'$  and analysis of sign of  $f'$  on domain

1 point is earned for  $(-\infty, \infty)$

$$f'(x) = 2 - 2x, x \leq 1$$

$$2 - 2x > 0$$

$$x < 1$$

$$f'(x) = 2x - 2, x > 1$$

$$2x - 2 > 0$$

**2-4-5**

$$x > 1$$

Since  $f$  increases on each of  $(-\infty, 1)$  and  $(1, \infty)$  and is continuous at  $x = 1$ ,  $f$  is increasing on  $(-\infty, \infty)$ .

**Part C**

1 point is earned for  $f''$

1 point is earned for noting sign changes in  $f''$  or change of concavity in  $f$

1 point is earned for ordered pair (or equivalent)

$$f''(x) = -2, x < 1$$

$$f''(x) = 2, x > 1$$

Since  $f''(x) < 0$  on  $(-\infty, 1)$  and

$f''(x) > 0$  on  $(1, \infty)$  and

$f(1)$  is defined,

$(1, f(1)) = (1, 1)$  is a point of inflection.



|   |   |   |   |
|---|---|---|---|
| 0 | 1 | 2 | 3 |
|---|---|---|---|

The student response earns all of the following points:

1 point is earned for  $f''$

1 point is earned for noting sign changes in  $f''$  or change of concavity in  $f$

1 point is earned for ordered pair (or equivalent)

$$f''(x) = -2, x < 1$$

$$f''(x) = 2, x > 1$$

Since  $f''(x) < 0$  on  $(-\infty, 1)$  and

$f''(x) > 0$  on  $(1, \infty)$  and

$f(1)$  is defined,

$(1, f(1)) = (1, 1)$  is a point of inflection.

## 2-4-5

12. Let  $f$  be a function defined by  $f(x) = \begin{cases} 2x - x^2 & \text{for } x \leq 1, \\ x^2 + kx + p & \text{for } x > 1. \end{cases}$   
For what values of  $k$  and  $p$  will  $f$  be continuous and differentiable at  $x=1$ ?

## Part A

1 point is earned for equation of  $f(1)$

1 point is earned for  $1 = 1 + k + p$

1 point is earned for  $0 = 2 + k$

1 point is earned for solution of system

For continuity at  $x = 1$ ,

$$\lim_{x \rightarrow 1^-} (2x - x^2) = f(1) = \lim_{x \rightarrow 1^+} (x^2 + kx + p)$$

Therefore  $1 = 1 + k + p$

Since  $f$  is continuous at  $x=1$  and is piecewise polynomial, left and right derivatives exist.

$$f'_-(1) = 0 \text{ and } f'_+(1) = 2 + k.$$

For differentiability at  $x = 1, 0 = 2 + k$ .

Therefore  $k = -2, p = 2$



|   |   |   |   |   |
|---|---|---|---|---|
| 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|

The student response earns four of the following points:

1 point is earned for equation of  $f(1)$

1 point is earned for  $1 = 1 + k + p$

1 point is earned for  $0 = 2 + k$

1 point is earned for solution of system

For continuity at  $x = 1$ ,

$$\lim_{x \rightarrow 1^-} (2x - x^2) = f(1) = \lim_{x \rightarrow 1^+} (x^2 + kx + p)$$

**2-4-5**

Therefore  $1 = 1 + k + p$

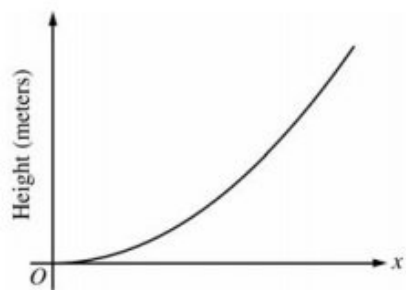
Since  $f$  is continuous at  $x=1$  and is piecewise polynomial, left and right derivatives exist.

$$f'_-(1) = 0 \text{ and } f'_+(1) = 2 + k.$$

For differentiability at  $x = 1, 0 = 2 + k$ .

Therefore  $k = -2, p = 2$

2-4-5



The figure above is the graph of a function of  $x$ , which models the height of a skateboard ramp. The function meets the following requirements.

- (i) At  $x=0$ , the value of the function is 0, and the slope of the graph of the function is 0.
- (ii) At  $x=4$ , the value of the function is 1, and the slope of the graph of the function is 1.
- (iii) Between  $x=0$  and  $x=4$ , the function is increasing.

13. Let  $f(x) = ax^2$ , where  $a$  is a nonzero constant. Show that it is not possible to find a value for  $a$  so that  $f$  meets requirement (ii) above.

### Part A

1 point is earned for if  $a = \frac{1}{16}$  or  $a = \frac{1}{8}$

1 point is earned if shows  $a$  does not work

$f(4)=1$  implies that  $a = \frac{1}{16}$  and  $f'(4)=2a(4)=1$  implies that  $a = \frac{1}{8}$ . Thus,  $f$  cannot satisfy (ii).



|   |   |   |
|---|---|---|
| 0 | 1 | 2 |
|---|---|---|

The student response earns all of the following points:

1 point is earned for if  $a = \frac{1}{16}$  or  $a = \frac{1}{8}$

1 point is earned if shows  $a$  does not work

$f(4)=1$  implies that  $a = \frac{1}{16}$  and  $f'(4)=2a(4)=1$  implies that  $a = \frac{1}{8}$ . Thus,  $f$  cannot satisfy (ii).

## 2-4-5

14. Let  $g(x) = cx^3 - \frac{x^2}{16}$  where  $c$  is a nonzero constant. Find the value of  $c$  so that  $g$  meets requirement (ii) above. Show the work that leads to your answer.

## Part B

1 point is earned for the value of  $c$

$$g(4) = 64c - 1 = 1 \text{ implies that } c = \frac{1}{32}.$$

$$\text{When } c = \frac{1}{32}, g'(4) = 3c(4)^2 - \frac{2(4)}{16} = 3\left(\frac{1}{32}\right)(16) - \frac{1}{2} = 1$$



|   |   |
|---|---|
| 0 | 1 |
|---|---|

The student response earns none of the following points:

1 point is earned for the value of  $c$

$$g(4) = 64c - 1 = 1 \text{ implies that } c = \frac{1}{32}.$$

$$\text{When } c = \frac{1}{32}, g'(4) = 3c(4)^2 - \frac{2(4)}{16} = 3\left(\frac{1}{32}\right)(16) - \frac{1}{2} = 1$$

15. Let  $f$  be the function given by  $f(x) = \sqrt{x^4 - 16x^2}$ . Find  $f'(x)$ .

## Part C

The response can earn up to 3 points:

3 points: For the correct derivative

$$f'(x) = \frac{1}{2}(x^4 - 16x^2)^{-\frac{1}{2}}(4x^3 - 32x) = \frac{2x(x^2 - 8)}{|x|\sqrt{x^2 - 16}}$$



|   |   |   |   |
|---|---|---|---|
| 0 | 1 | 2 | 3 |
|---|---|---|---|

The response earns all three of the following points:

3 points: For the correct derivative

$$f'(x) = \frac{1}{2}(x^4 - 16x^2)^{-\frac{1}{2}}(4x^3 - 32x) = \frac{2x(x^2 - 8)}{|x|\sqrt{x^2 - 16}}$$

## 2-4-5

16.

|                    |   |     |     |      |      |      |      |
|--------------------|---|-----|-----|------|------|------|------|
| $t$<br>(minutes)   | 0 | 1   | 2   | 3    | 4    | 5    | 6    |
| $C(t)$<br>(ounces) | 0 | 5.3 | 8.8 | 11.2 | 12.8 | 13.8 | 14.5 |

Hot water is dripping through a coffeemaker, filling a large cup with coffee. The amount of coffee in the cup at time  $t$ ,  $0 \leq t \leq 6$ , is given by a differentiable function  $C$ , where  $t$  is measured in minutes. Selected values of  $C(t)$ , measured in ounces, are given in the table above.

Is there a time  $t$ ,  $2 \leq t \leq 4$ , at which  $C'(t) = 2$ ? Justify your answer.

## Part B

1 point is earned for  $\frac{C(4) - C(2)}{4 - 2}$

1 point is earned for the conclusion, using MVT

$C$  is differentiable  $\Rightarrow C$  is continuous (on the closed interval)

$$\frac{C(4) - C(2)}{4 - 2} = \frac{12.8 - 8.8}{2} = 2$$

Therefore, by the Mean Value Theorem, there is at least one time  $t$ ,  $2 < t < 4$ , for which  $C'(t) = 2$ .



|   |   |   |
|---|---|---|
| 0 | 1 | 2 |
|---|---|---|

The student response earns all of the following points:

1 point is earned for  $\frac{C(4) - C(2)}{4 - 2}$

1 point is earned for the conclusion, using MVT

$C$  is differentiable  $\Rightarrow C$  is continuous (on the closed interval)

$$\frac{C(4) - C(2)}{4 - 2} = \frac{12.8 - 8.8}{2} = 2$$

Therefore, by the Mean Value Theorem, there is at least one time  $t$ ,  $2 < t < 4$ , for which  $C'(t) = 2$ .

17. Let  $f$  be the function defined by  $f(x) = \sqrt{|x - 2|}$  for all  $x$ . Which of the following statements is true?

(A)  $f$  is continuous but not differentiable at  $x = 2$ .



(B)  $f$  is differentiable at  $x = 2$ .

(C)  $f$  is not continuous at  $x = 2$ .

(D)  $\lim_{x \rightarrow 2} f(x) \neq 0$

(E)  $x = 2$  is a vertical asymptote of the graph of  $f$ .

2-4-5

18. 
$$f(x) = \begin{cases} x + 2 & \text{if } x \leq 3 \\ 4x - 7 & \text{if } x > 3 \end{cases}$$

Let  $f$  be the function given above. Which of the following statements are true about  $f$ ?

I.  $\lim_{x \rightarrow 3} f(x)$  exists.

II.  $f$  is continuous at  $x = 3$ .

III.  $f$  is differentiable at  $x = 3$ .

(A) None

(B) I only

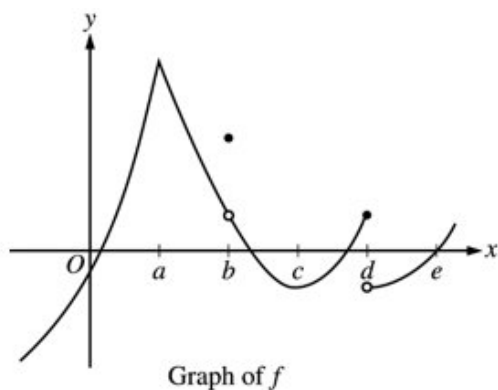
(C) II only

(D) I and II only

(E) I, II, and III



19.



The graph of a function  $f$  is shown above. At which value of  $x$  is  $f$  continuous, but not differentiable?

(A)  $a$

(B)  $b$

(C)  $c$

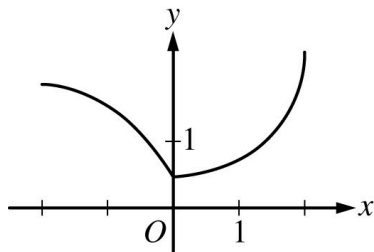
(D)  $d$

(E)  $e$



2-4-5

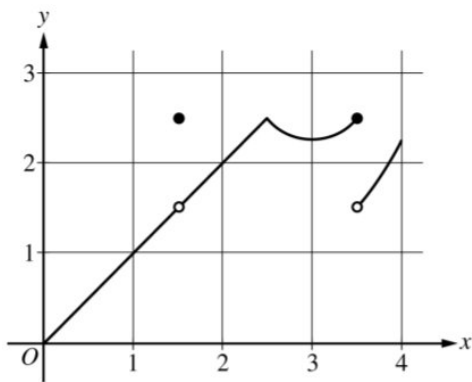
20.

Graph of  $f$ 

The function  $f$ , whose graph is shown above, is defined on the interval  $-2 \leq x \leq 2$ . Which of the following statements about  $f$  is false?

- (A)  $f$  is continuous at  $x = 0$ .
- (B)  $f$  is differentiable at  $x = 0$ . ✓
- (C)  $f$  has a critical point at  $x = 0$ .
- (D)  $f$  has an absolute minimum at  $x = 0$ .
- (E) The concavity of the graph of  $f$  changes at  $x = 0$ .

21.

Graph of  $f$ 

The graph of the function  $f$  is shown above.

Of the following intervals, on which is  $f$  continuous but not differentiable?

- (A)  $(0, 1)$
- (B)  $(1, 2)$
- (C)  $(2, 3)$  ✓
- (D)  $(3, 4)$

## 2-4-5

22.

I. If  $f$  is defined at  $x = a$ , then  $\lim_{x \rightarrow a} f(x) = f(a)$ .II. If  $f$  is continuous at  $x = a$ , then  $\lim_{x \rightarrow a} f(x) = f(a)$ .III. If  $f$  is differentiable at  $x = a$ , then  $\lim_{x \rightarrow a} f(x) = f(a)$ .For any function  $f$ , which of the following statements must be true?

(A) III only

(B) I and II only

(C) II and III only

(D) I, II, and III



23.

If  $f(x) = 1 + x^{\frac{2}{3}}$ , which of the following is NOT true?(A)  $f$  is continuous for all real numbers.(B)  $f$  has a minimum at  $x=0$ .(C)  $f$  is increasing for  $x>0$ .(D)  $f'(x)$  exists for all  $x$ .(E)  $f''(x)$  is negative for  $x>0$ .

24.

If  $g$  is the function given by  $g(x) = \frac{1}{3}x^3 + \frac{3}{2}x^2 - 70x + 5$ , on which of the following intervals is  $g$  decreasing?(A)  $(-\infty, -10)$  and  $(7, \infty)$ (B)  $(-\infty, -7)$  and  $(10, \infty)$ (C)  $(-\infty, 10)$ (D)  $(-10, 7)$ (E)  $(-7, 10)$ 

25.

If  $f(x) = 2 + |x-3|$  for all  $x$ , then the value of the derivative  $f'(x)$  at  $x = 3$  is

(A) -1

(B) 0

(C) 1

(D) 2

(E) Nonexistent



26.

$$f(x) = \begin{cases} 3x + 5 & \text{when } x < -1 \\ -x^2 + 3 & \text{when } x \geq -1 \end{cases}$$

If  $f$  is the function defined above, then  $f'(-1)$  is

(A) -3

(B) -2

(C) 2

(D) 3

(E) nonexistent



## 2-4-5

27. Which of the following is an equation of a curve that intersects at right angles every curve of the family  $y = \frac{1}{x} + k$  (where  $k$  takes all real values)?

(A)  $y = -x$   
(B)  $y = -x^2$   
(C)  $y = -\frac{1}{3}x^3$   
(D)  $y = \frac{1}{3}x^3$   
(E)  $y = \ln x$



28. If  $f(x) = x$ , then  $f'(5) =$

(A) 0  
(B)  $1/5$   
(C) 1  
(D) 5  
(E)  $25/2$



29. If  $f(x) = \sqrt{x} + \frac{3}{\sqrt{x}}$ , then  $f'(4) =$

(A)  $\frac{1}{16}$   
(B)  $\frac{5}{16}$   
(C) 1  
(D)  $\frac{7}{2}$   
(E)  $\frac{49}{4}$



30. If  $f(x) = x^{\frac{3}{2}}$ , then  $f(4) =$

(A) -6  
(B) -3  
(C) 3  
(D) 6  
(E) 8



31. 
$$f(x) = \begin{cases} x + b & \text{if } x \leq 1 \\ ax^2 & \text{if } x > 1 \end{cases}$$

Let  $f$  be the function given above. What are all values of  $a$  and  $b$  for which  $f$  is differentiable at  $x = 1$ ?

(A)  $a = \frac{1}{2}$  and  $b = -\frac{1}{2}$   
(B)  $a = \frac{1}{2}$  and  $b = \frac{3}{2}$   
(C)  $a = \frac{1}{2}$  and  $b$  is any real number  
(D)  $a = b + 1$ , where  $b$  is any real number  
(E) There are no such values of  $a$  and  $b$ .

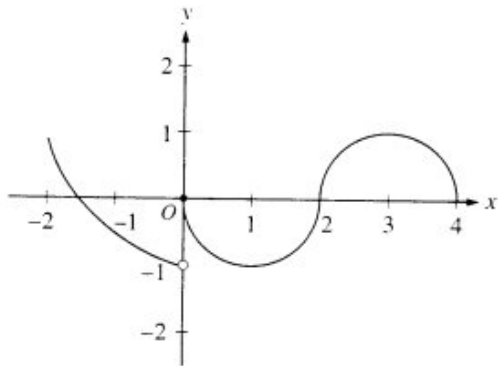


## 2-4-5

32. If  $f$  is a continuous function on  $[a, b]$ , which of the following is necessarily true?

- (A)  $f'$  exists on  $(a, b)$
- (B) If  $f(x_0)$  is a maximum of  $f$ , then  $f'(x_0) = 0$ .
- (C)  $\lim_{x \rightarrow x_0} f(x) = f\left(\lim_{x \rightarrow x_0} x\right)$  for  $x_0 \in (a, b)$  ✓
- (D)  $f'(x) = 0$  for some  $x \in [a, b]$
- (E) The graph of  $f'$  is a straight line.

33.



The graph of the function  $f$  shown in the figure above has a vertical tangent at the point  $(2, 0)$  and horizontal tangents at the points  $(1, -1)$  and  $(3, 1)$ . For what values of  $x$ ,  $-2 < x < 4$ , is  $f$  not differentiable?

- (A) 0 only
- (B) 0 and 2 only ✓
- (C) 1 and 3 only
- (D) 0, 1, and 3 only
- (E) 0, 1, 2, and 3

34. 
$$f(x) = \begin{cases} 2x + 5 & \text{for } x < -1 \\ -x^2 + 6 & \text{for } x \geq -1 \end{cases}$$

If  $f$  is the function defined above, then  $f'(-1)$  is

- (A) -2
- (B) 2
- (C) 3
- (D) 5
- (E) nonexistent ✓

## 2-4-5

35. 
$$f(x) = \begin{cases} cx + d & \text{for } x \leq 2 \\ x^2 - cx & \text{for } x > 2 \end{cases}$$

Let  $f$  be the function defined above, where  $c$  and  $d$  are constants. If  $f$  is differentiable at  $x = 2$  what is the value of  $c + d$ ?

(A)  $-4$

(B)  $-2$  ✓

(C)  $0$

(D)  $2$

(E)  $4$

36. 
$$f(x) = \begin{cases} cx + d & \text{for } x \leq 2 \\ x^2 - cx & \text{for } x > 2 \end{cases}$$

Let  $f$  be the function defined above, where  $c$  and  $d$  are constants. If  $f$  is differentiable at  $x = 2$ , what is the value of  $c + d$ ?

(A)  $-4$

(B)  $-2$  ✓

(C)  $0$

(D)  $2$

(E)  $4$

37. The function  $g$  is differentiable at  $x = 5$ , but its derivative  $g'$  is not continuous at  $x = 5$ . Which of the following statements must be true?

(A)  $g$  is continuous at  $x = 5$ . ✓

(B)  $g'$  is differentiable at  $x = 5$ .

(C)  $g$  has a critical point at  $x = 5$ .

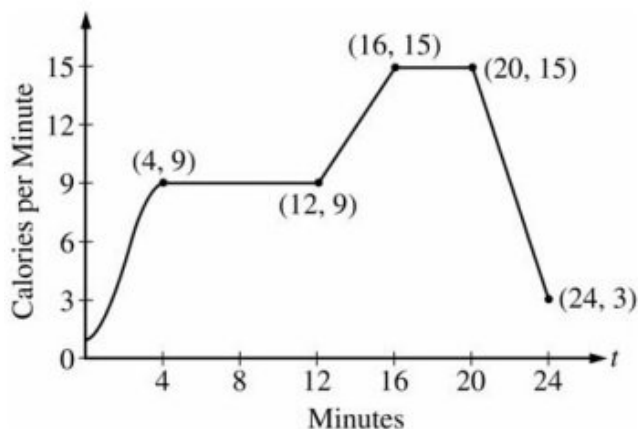
(D) The graph of  $g$  has a vertical tangent line at  $x = 5$ .

**Answer A**

Correct. This statement is true. Since  $g$  is differentiable at  $x = 5$ ,  $g$  is continuous at  $x = 5$ .

2-4-5

38.



The rate, in calories per minute, at which a person using an exercise machine burns calories is modeled by the function  $f$ . In the figure above,  $f(t) = -\frac{1}{4}t^3 + \frac{3}{2}t^2 + 1$  for  $0 \leq t \leq 4$  and  $f$  is piecewise linear for  $4 \leq t \leq 24$ .

For the time interval  $0 \leq t \leq 24$ , at what time  $t$  is  $f$  increasing at its greatest rate? Show the reasoning that supports your answer.

**Part B**

1 point is earned for  $f$  on (0,4)

1 point is earned for showing  $f'$  has a max at  $t = 2$  on (0,4)

1 point is earned if shows  $12 < t < 16, f'(t) < f'(2)$

1 point is earned for the answer

$f$  is increasing on  $[0, 4]$  and on  $[12, 16]$ .

On  $(12, 16)$ ,  $f'(t) = \frac{15-9}{16-12} = \frac{3}{2}$  since  $f$  has constant slope on this interval.

On  $(0, 4)$ ,  $f'(t) = -\frac{3}{4}t^2 + 3t$  and  $f''(t) = -\frac{3}{2}t + 3 = 0$  when  $t = 2$ . This is where  $f$  has a maximum on  $[0, 4]$  since  $f'' > 0$  on  $(0, 2)$  and  $f'' < 0$  on  $(2, 4)$ .

On  $[0, 24]$ ,  $f$  is increasing at its greatest rate when  $t = 2$  because  $f'(2) = 3 > \frac{3}{2}$ .



|   |   |   |   |   |
|---|---|---|---|---|
| 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|

The student response earns all of the following points:

1 point is earned for  $f$  on (0,4)

1 point is earned for showing  $f'$  has a max at  $t = 2$  on (0,4)

1 point is earned if shows  $12 < t < 16, f'(t) < f'(2)$

1 point is earned for the answer

**2-4-5**

$f$  is increasing on  $[0, 4]$  and on  $[12, 16]$ .

On  $(12, 16)$ ,  $f'(t) = \frac{15-9}{16-12} = \frac{3}{2}$  since  $f$  has constant slope on this interval.

On  $(0, 4)$ ,  $f'(t) = -\frac{3}{4}t^2 + 3t$  and  $f''(t) = -\frac{3}{2}t + 3 = 0$  when  $t = 2$ . This is where  $f$  has a maximum on  $[0, 4]$  since  $f' > 0$  on  $(0, 2)$  and  $f' < 0$  on  $(2, 4)$ .

On  $[0, 24]$ ,  $f$  is increasing at its greatest rate when  $t = 2$  because  $f'(2) = 3 > \frac{3}{2}$ .

**39.** If  $f$  is a differentiable function such that  $f(3)=8$  and  $f'(3) = 5$ , which of the following statements could be false?

(A)  $\lim_{x \rightarrow 3} f(x) = 8$

(B)  $\lim_{x \rightarrow 3^+} f(x) = \lim_{x \rightarrow 3^-} f(x)$

(C)  $\lim_{x \rightarrow 3} \frac{f(x)-8}{x-3} = 5$

(D)  $\lim_{h \rightarrow 0} \frac{f(3+h)-8}{h} = 5$

(E)  $\lim_{x \rightarrow 3} f'(x) = 5$



2-4-5

40.

|                          |   |     |    |      |      |
|--------------------------|---|-----|----|------|------|
| $t$ (minutes)            | 0 | 2   | 5  | 8    | 12   |
| $v_A(t)$ (meters/minute) | 0 | 100 | 40 | -120 | -150 |

Train  $A$  runs back and forth on an east-west section of railroad track. Train  $A$ 's velocity, measured in meters per minute, is given by a differentiable function  $v_A(t)$ , where time  $t$  is measured in minutes. Selected values for  $v_A(t)$  are given in the table above.

Do the data in the table support the conclusion that train  $A$ 's velocity is -100 meters per minute at some time  $t$  with  $5 < t < 8$ ? Give a reason for your answer.

**Part B**

1 point is earned for  $V_A(8) < -100 < V_A(5)$

1 point is earned for the conclusion, using IVT

$V_A$  is differentiable  $\Rightarrow V_A$  is continuous

$V_A(8) = -120 < -100 < 40 = V_A(5)$

Therefore, by the Intermediate Value Theorem, there is a time  $t$ ,  $5 < t < 8$ , such that  $V_A(t) = -100$ .



|   |   |   |
|---|---|---|
| 0 | 1 | 2 |
|---|---|---|

The student response earns two of the following points:

1 point is earned for  $V_A(8) < -100 < V_A(5)$

1 point is earned for the conclusion, using IVT

$V_A$  is differentiable  $\Rightarrow V_A$  is continuous

$V_A(8) = -120 < -100 < 40 = V_A(5)$  Therefore, by the Intermediate Value Theorem, there is a time  $t$ ,  $5 < t < 8$ , such that  $V_A(t) = -100$ .

41.

$$f(x) = \begin{cases} \frac{x^2-4}{x-2} & \text{if } x \neq 2 \\ 1 & \text{if } x = 2 \end{cases}$$

Let  $f$  be the function defined above. Which of the following statements about  $f$  are true?

I.  $f$  has a limit at  $x = 2$ .

II.  $f$  is continuous at  $x = 2$ .

III.  $f$  is differentiable at  $x = 2$ .

(A) I only

(B) II only

(C) III only

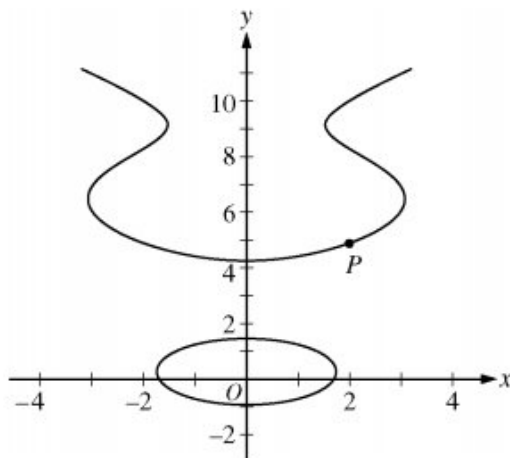
(D) I and II only

(E) I, II, and III



2-4-5

42.



The graph of the equation  $x^2 = -2 + y + 5 \cos y$  is shown above for  $y \leq 11$ . It is known that  $\frac{dy}{dx} = \frac{2x}{1-5 \sin y}$ . The x-coordinate of point  $P$  shown on the graph is 2.

 For  $y \leq 11$ , find the y-coordinate of each point on the graph where the line tangent to the graph at that point is vertical.

**General**

**1 point is earned for:** sets  $1 - 5 \sin y = 0$

**2 points are earned for:** y-coordinate

$$1 - 5 \sin y = 0$$

$$y = 0.201, 6.485 \text{ (or } 6.484), 9.223$$



|   |   |   |   |
|---|---|---|---|
| 0 | 1 | 2 | 3 |
|---|---|---|---|

The student response earns all of the following points:

**1 point is earned for:** sets  $1 - 5 \sin y = 0$

**2 points are earned for:** y-coordinate

$$1 - 5 \sin y = 0$$

$$y = 0.201, 6.485 \text{ (or } 6.484), 9.223$$

## 2-4-5

43. If  $f(x) = x^3 - x^2 + x - 1$ , then  $f'(2) =$

(A) 10

(B) 9

(C) 7

(D) 5

(E) 3



44. If  $f$  is the function given by  $f(x) = \frac{4}{x} + 5x - 1$ , then  $f'(2) =$

(A) 4

(B) 6

(C) 7

(D) 11

**Answer A**

45. Which of the following statements about the function  $f$ , if true, cannot be used to conclude that  $f$  is defined at  $x = 1$ ?

(A)  $\lim_{x \rightarrow 1} f(x)$  exists.

(B)  $f$  is continuous at  $x = 1$ .

(C)  $f$  is differentiable at  $x = 1$ .

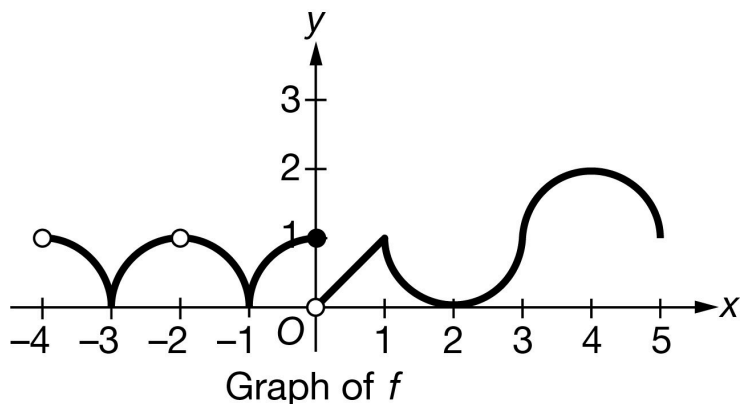
(D) The line tangent to the graph of  $f$  at  $x = 1$  exists.

**Answer A**

Correct. The existence of  $\lim_{x \rightarrow 1} f(x)$  does not guarantee that  $f$  is defined at  $x = 1$ . For example,  $f(x) = \frac{(x-1)^3}{x-1}$  is not defined at  $x = 1$ , but  $\lim_{x \rightarrow 1} f(x)$  exists.

2-4-5

46.

Graph of  $f$ 

The graph of the function  $f$ , shown above, has a vertical tangent at  $x = 3$  and horizontal tangents at  $x = 2$  and  $x = 4$ . Which of the following statements is false?

- (A)  $f$  is not differentiable at  $x = 3$  because the graph of  $f$  has a vertical tangent at  $x = 3$ .
- (B)  $f$  is not differentiable at  $x = -2$  and  $x = 0$  because  $f$  is not continuous at  $x = -2$  and  $x = 0$ .
- (C)  $f$  is not differentiable at  $x = -1$  and  $x = 1$  because the graph of  $f$  has sharp corners at  $x = -1$  and  $x = 1$ .
- (D)  $f$  is not differentiable at  $x = 2$  and  $x = 4$  because the graph of  $f$  has horizontal tangents at  $x = 2$  and  $x = 4$ . ✓

**Answer D**

Correct. A function is differentiable at points at which the line tangent to the graph is horizontal.

## 2-4-5

47.

$$f(x) = \begin{cases} 4 - x^2 & \text{for } x \leq 1 \\ 5 - 2x & \text{for } x > 1 \end{cases}$$

Let  $f$  be the function defined as shown. Which of the following statements about  $f$  is true at  $x = 1$ ?

- (A)  $f$  is both continuous and differentiable. ✓
- (B)  $f$  is continuous but not differentiable.
- (C)  $f$  is differentiable but not continuous.
- (D)  $f$  is neither continuous nor differentiable.

**Answer A**

Correct. This statement is true.

$$\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^-} (4 - x^2) = 4 - 1 = 3$$

$$\lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^+} (5 - 2x) = 5 - 2 = 3$$

Therefore,  $\lim_{x \rightarrow 1} f(x)$  exists and  $\lim_{x \rightarrow 1} f(x) = 3 = f(1)$ , so  $f$  is continuous at  $x = 1$ .

In addition,

$$\lim_{h \rightarrow 0^-} \frac{f(1+h) - f(1)}{h} = \lim_{h \rightarrow 0^-} \frac{4 - (1+h)^2 - 3}{h} = \lim_{h \rightarrow 0^-} \frac{-2h - h^2}{h} = -2$$

$$\lim_{h \rightarrow 0^+} \frac{f(1+h) - f(1)}{h} = \lim_{h \rightarrow 0^+} \frac{5 - 2(1+h) - 3}{h} = \lim_{h \rightarrow 0^+} \frac{-2h}{h} = -2$$

Therefore,  $f$  is also differentiable at  $x = 1$  and  $f'(1) = -2$ .

An alternative way to see that the piecewise defined function  $f$  is differentiable at  $x = 1$  is to observe that

$$f'(x) = \begin{cases} -2x & \text{for } x < 1 \\ -2 & \text{for } x > 1 \end{cases}$$

Since  $f$  is continuous at  $x = 1$  and the derivatives  $-2x$  and  $-2$  are equal at  $x = 1$ , then  $f$  is differentiable at  $x = 1$ . For details on this method, see the article “Differentiability of Piecewise Defined Functions” in the Classroom Resources section of the Calculus AB or Calculus BC course page at AP Central.

## 2-4-5

48.

$$f(x) = \begin{cases} x^2 - 4x & \text{for } x < 2 \\ x^3 - 12 & \text{for } x \geq 2 \end{cases}$$

Let  $f$  be the function defined as shown. Which of the following statements about  $f$  is false?

- (A)  $\lim_{x \rightarrow 2^-} f(x)$  exists.
- (B)  $\lim_{x \rightarrow 2} f(x)$  exists.
- (C)  $f$  is continuous at  $x = 2$ .
- (D)  $f$  is differentiable at  $x = 2$ . ✓

**Answer D**

Correct. This statement is false.

$$\begin{aligned} \lim_{h \rightarrow 0^-} \frac{f(2+h) - f(2)}{h} &= \lim_{h \rightarrow 0^-} \frac{(2+h)^2 - 4(2+h) - (-4)}{h} = \lim_{h \rightarrow 0^-} \frac{h^2}{h} = \lim_{h \rightarrow 0^-} h = 0 \\ \lim_{h \rightarrow 0^+} \frac{f(2+h) - f(2)}{h} &= \lim_{h \rightarrow 0^+} \frac{(2+h)^3 - 12 - (-4)}{h} = \lim_{h \rightarrow 0^+} \frac{h^3 + 6h^2 + 12h}{h} = 12 \end{aligned}$$

Therefore,  $f$  is not differentiable at  $x = 2$  because the one-sided limits are not equal.

An alternative way to see that the piecewise defined function  $f$  is not differentiable at  $x = 2$  is to observe that

$$f'(x) = \begin{cases} 2x - 4 & \text{for } x < 2 \\ 3x^2 & \text{for } x > 2 \end{cases}$$

Since the derivatives  $2x - 4$  and  $3x^2$  are not equal at  $x = 2$ , then  $f$  is not differentiable at

$x = 2$ . For details on this method, see the article “Differentiability of Piecewise Defined Functions” in the Classroom Resources section of the Calculus AB or Calculus BC course page at AP Central.

2-4-5

49.

$$f(x) = \begin{cases} x^2 - 2x & \text{for } x \leq 4 \\ g(x) & \text{for } x > 4 \end{cases}$$

Let  $f$  be the function defined as shown. If  $f$  is differentiable, which of the following could define  $g(x)$ ?

(A)  $6x$

(B)  $x^{\frac{3}{2}}$

(C)  $x^2 - 8$

(D)  $\frac{1}{8}x^3$

**Answer D**

Correct. For  $f$  to be differentiable at  $x = 4$ , two conditions must hold for a differentiable function  $g(x)$ .

(1)  $g(4) = (x^2 - 2x)|_{x=4} = 16 - 8 = 8$  (continuity at  $x = 4$ )

(2)  $g'(4) = \frac{d}{dx}(x^2 - 2x)|_{x=4} = (2x - 2)|_{x=4} = 6$  (differentiability at  $x = 4$ )

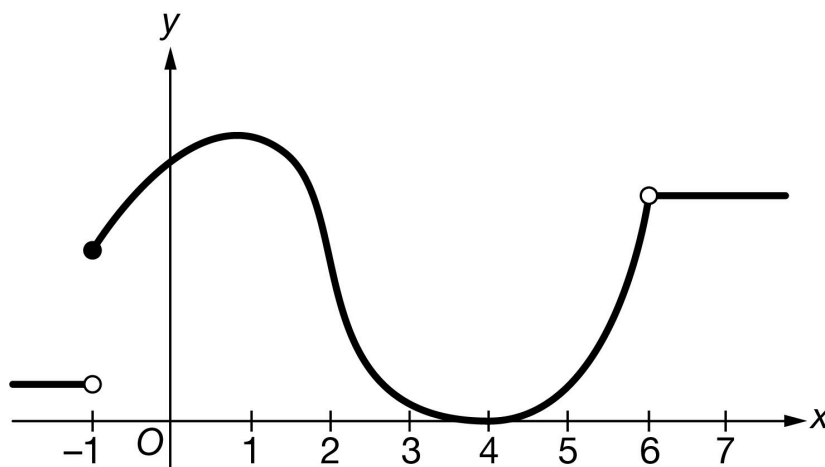
The differentiable function  $g(x) = \frac{1}{8}x^3$  satisfies both conditions.

(For more information, see “Differentiability of Piecewise Defined Functions,”

<https://apcentral.collegeboard.org/courses/resources/differentiability-piecewise-defined-functions>, in the Classroom Resources section of the AB/BC courses at AP Central.)

2-4-5

50.

Graph of  $f$ 

The figure above shows the graph of a function  $f$ , which has a vertical tangent at  $x = 2$  and a horizontal tangent at  $x = 4$ . Which of the following statements is false?

- (A)  $f$  is not differentiable at  $x = -1$  because the graph of  $f$  has a jump discontinuity at  $x = -1$ .
- (B)  $f$  is not differentiable at  $x = 2$  because the graph of  $f$  has a vertical tangent at  $x = 2$ .
- (C)  $f$  is not differentiable at  $x = 4$  because the graph of  $f$  has a horizontal tangent at  $x = 4$ . ✓
- (D)  $f$  is not differentiable at  $x = 6$  because the graph of  $f$  has a removable discontinuity at  $x = 6$ .

**Answer C**

Correct. A function is differentiable at points at which the line tangent to the graph is horizontal.

51.

$$f(x) = \begin{cases} x^2 - 20 & \text{for } x < 5 \\ -x^2 + 20 & \text{for } x \geq 5 \end{cases}$$

Let  $f$  be the function defined above. Which of the following statements is true?

- (A)  $f$  is not differentiable at  $x = 5$  because  $f$  is not continuous at  $x = 5$ . ✓
- (B)  $f$  is not differentiable at  $x = 5$  because the graph of  $f$  has a sharp corner at  $x = 5$ .
- (C)  $f$  is not differentiable at  $x = 5$  because the graph of  $f$  has a vertical tangent at  $x = 5$ .
- (D)  $f$  is not differentiable at  $x = 5$  because  $f$  is not defined at  $x = 5$ .

**Answer A**

Correct. The limits  $\lim_{x \rightarrow 5^-} f(x) = \lim_{x \rightarrow 5^-} x^2 - 20 = 5^2 - 20 = 5$  and  $\lim_{x \rightarrow 5^+} f(x) = \lim_{x \rightarrow 5^+} -x^2 + 20 = -(5^2) + 20 = -5$  are not equal. Therefore,  $f$  is not continuous at  $x = 5$ . If a function is not continuous at a point, then it is not differentiable at that point.

## 2-4-5

52. Which of the following statements, if true, can be used to conclude that  $f(2)$  exists?

- I.  $\lim_{x \rightarrow 2} f(x)$  exists.
- II.  $f$  is continuous at  $x = 2$ .
- III.  $f$  is differentiable at  $x = 2$ .

(A) I only

(B) II only

(C) II and III only

(D) I, II, and III

**Answer C**

Correct. If  $f$  is continuous at  $x = 2$ , then  $f(2)$  exists. Also if  $f$  is differentiable at  $x = 2$ , then  $f$  is continuous at  $x = 2$  and  $f(2)$  exists.

53.

$$f(x) = \begin{cases} 3x + 1 & \text{for } x \leq 2 \\ 5x - 3 & \text{for } x > 2 \end{cases}$$

Let  $f$  be the function defined above. Which of the following statements is true?

(A)  $f$  is neither continuous nor differentiable at  $x = 2$ .

(B)  $f$  is continuous but not differentiable at  $x = 2$ .

(C)  $f$  is differentiable but not continuous at  $x = 2$ .

(D)  $f$  is both continuous and differentiable at  $x = 2$ .

**Answer B**

Correct.  $f$  is continuous at  $x = 2$  because  $\lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^-} (3x + 1) = 7$ ,  $\lim_{x \rightarrow 2^+} f(x) = \lim_{x \rightarrow 2^+} (5x - 3) = 7$ , and  $f(2) = 7$ .

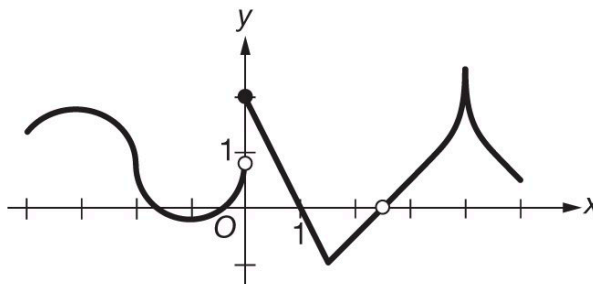
$f$  is not differentiable at  $x = 2$  because the left-hand derivative does not equal the right-hand derivative at  $x = 2$ , as follows.

$$\lim_{h \rightarrow 0^-} \frac{f(2+h) - f(2)}{h} = \lim_{h \rightarrow 0^-} \frac{(3(2+h) + 1) - 7}{h} = \lim_{h \rightarrow 0^-} \frac{3h}{h} = 3$$

$$\lim_{h \rightarrow 0^+} \frac{f(2+h) - f(2)}{h} = \lim_{h \rightarrow 0^+} \frac{(5(2+h) - 3) - 7}{h} = \lim_{h \rightarrow 0^+} \frac{5h}{h} = 5$$

2-4-5

54.

Graph of  $f$ 

The graph of the function  $f$ , shown above, has a vertical tangent at  $x = -2$  and horizontal tangents at  $x = -3$  and  $x = -1$ . Which of the following statements is false?

- (A)  $f$  is not differentiable at  $x = -2$  because the graph of  $f$  has a vertical tangent at  $x = -2$ .
- (B)  $f$  is not differentiable at  $x = 0$  and  $x = 2.5$  because  $f$  is not continuous at  $x = 0$  and  $x = 2.5$ .
- (C)  $f$  is not differentiable at  $x = 1.5$  and  $x = 4$  because the graph of  $f$  has sharp corners at  $x = 1.5$  and  $x = 4$ .
- (D)  $f$  is not differentiable at  $x = -3$  and  $x = -1$  because the graph of  $f$  has horizontal tangents at  $x = -3$  and  $x = -1$ . ✓

**Answer D**

Correct. A function is differentiable at points at which the line tangent to the graph is horizontal.

55.

$$f(x) = \begin{cases} \frac{2}{x} & \text{for } x < -1 \\ x^2 - 3 & \text{for } -1 \leq x \leq 2 \\ 4x - 3 & \text{for } x > 2 \end{cases}$$

Let  $f$  be the function defined above. At what values of  $x$ , if any, is  $f$  not differentiable?

- (A)  $x = -1$  only
- (B)  $x = 2$  only ✓
- (C)  $x = -1$  and  $x = 2$
- (D)  $f$  is differentiable for all values of  $x$ .

**Answer B**

The function  $f$  is not continuous at  $x = 2$  and so cannot be differentiable.

## 2-4-5

56.

$$f(x) = \begin{cases} 2 & \text{for } x < 5 \\ 2x - 4 & \text{for } x \geq 5 \end{cases}$$

Let  $f$  be the function defined above. Which of the following statements is true?

- (A)  $f$  is not differentiable at  $x = 5$  because  $f$  is not continuous at  $x = 5$ . ✓
- (B)  $f$  is not differentiable at  $x = 5$  because the graph of  $f$  has a sharp corner at  $x = 5$ .
- (C)  $f$  is not differentiable at  $x = 5$  because the graph of  $f$  has a vertical tangent at  $x = 5$ .
- (D)  $f$  is not differentiable at  $x = 5$  because  $f$  is not defined at  $x = 5$ .

**Answer A**

Correct. The limits  $\lim_{x \rightarrow 5^-} f(x) = \lim_{x \rightarrow 5^-} 2 = 2$  and  $\lim_{x \rightarrow 5^+} f(x) = \lim_{x \rightarrow 5^+} (2x - 4) = 6$  are not equal. Therefore  $f$  is not continuous at  $x = 5$ .

If a function is not continuous at a point, then it is not differentiable at that point.

57. Which of the following statements, if true, cannot be used to conclude that  $f(0)$  exists?

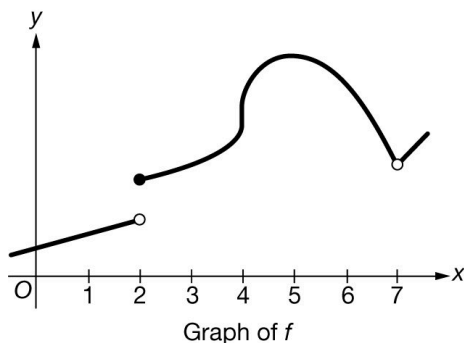
- (A)  $\lim_{x \rightarrow 0} f(x)$  exists. ✓
- (B)  $f$  is continuous at  $x = 0$ .
- (C)  $f$  is differentiable at  $x = 0$ .
- (D) The graph of  $f$  has a  $y$ -intercept.

**Answer A**

Correct. The existence of  $\lim_{x \rightarrow 0} f(x)$  does not guarantee that  $f(0)$  exists. For example,  $f(x) = \frac{x^2}{x}$  is not defined at  $x = 0$ , but  $\lim_{x \rightarrow 0} f(x)$  exists.

2-4-5

58.



Graph of  $f$

The figure above shows the graph of a function  $f$ , which has a vertical tangent at  $x = 4$  and a horizontal tangent at  $x = 5$ . Which of the following statements is false?

- (A)  $f$  is not differentiable at  $x = 2$  because the graph of  $f$  has a jump discontinuity at  $x = 2$ .
- (B)  $f$  is not differentiable at  $x = 4$  because the graph of  $f$  has a vertical tangent at  $x = 4$ .
- (C)  $f$  is not differentiable at  $x = 5$  because the graph of  $f$  has a horizontal tangent at  $x = 5$ . ✓
- (D)  $f$  is not differentiable at  $x = 7$  because the graph of  $f$  has a removable discontinuity at  $x = 7$ .

**Answer C**

Correct. A function is differentiable at points at which the line tangent to the graph is horizontal.

59. Which of the following statements, if true, can be used to conclude that  $f(3)$  exists?

- I.  $\lim_{x \rightarrow 3} f(x)$  exists.
- II.  $f$  is continuous at  $x = 3$ .
- III.  $f$  is differentiable at  $x = 3$ .
- (A) I only
- (B) II only
- (C) II and III only ✓
- (D) I, II, and III

**Answer C**

Correct. If  $f$  is continuous at  $x = 3$ , then  $f(3)$  exists. Also, if  $f$  is differentiable at  $x = 3$ , then  $f$  is continuous at  $x = 3$  and  $f(3)$  exists.

## 2-4-5

60.

$$f(x) = \begin{cases} 4x + 2 & \text{for } x \leq 2 \\ 3x + 4 & \text{for } x > 2 \end{cases}$$

Let  $f$  be the function defined above. Which of the following statements is true?

(A)  $f$  is neither continuous nor differentiable at  $x = 2$ .

(B)  $f$  is continuous but not differentiable at  $x = 2$ . ✓

(C)  $f$  is differentiable but not continuous at  $x = 2$ .

(D)  $f$  is both continuous and differentiable at  $x = 2$ .

**Answer B**

Correct.  $f$  is continuous at  $x = 2$  because  $\lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^-} (4x + 2) = 10$ ,  $\lim_{x \rightarrow 2^+} f(x) = \lim_{x \rightarrow 2^+} (3x + 4) = 10$ , and  $f(2) = 10$ .

$f$  is not differentiable at  $x = 2$  because the left-hand derivative does not equal the right-hand derivative at  $x = 2$ , as follows.

$$\begin{aligned} \lim_{h \rightarrow 0^-} \frac{f(2+h) - f(2)}{h} &= \lim_{h \rightarrow 0^-} \frac{(4(2+h) + 2) - 10}{h} = \lim_{h \rightarrow 0^-} \frac{4h}{h} = 4 \\ \lim_{h \rightarrow 0^+} \frac{f(2+h) - f(2)}{h} &= \lim_{h \rightarrow 0^+} \frac{(3(h+2) + 4) - 10}{h} = \lim_{h \rightarrow 0^+} \frac{3h}{h} = 3 \end{aligned}$$

## 2-4-5

## 61. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function  $f$  is assumed to be the set of all real numbers  $x$  for which  $f(x)$  is a real number.

$$f(t) = \begin{cases} 48t + t^2 - \frac{t^3}{12} & \text{for } 0 \leq t < 6 \\ g(t) & \text{for } 6 \leq t \leq 12 \end{cases}$$

|                       |     |     |     |     |
|-----------------------|-----|-----|-----|-----|
| $t$ (hours)           | 6   | 8   | 10  | 12  |
| $g(t)$ (cubic meters) | 306 | 376 | 428 | 474 |

At an excavation site, the amount of dirt that has been removed, in cubic meters, is modeled by the function  $f$  defined above, where  $g$  is a differentiable function and  $t$  is measured in hours. Values of  $g(t)$  at selected values of  $t$  are given in the table above.

- According to the model  $f$ , what is the average rate of change of the amount of dirt removed over the time interval  $6 \leq t \leq 12$  hours?
- Use the data in the table to approximate  $f'(9)$ , the instantaneous rate of change in the amount of dirt removed, in cubic meters per hour, at time  $t = 9$  hours. Show the computations that lead to your answer.
- Is  $f$  continuous for  $0 \leq t \leq 12$ ? Justify your answer.
- Find  $f'(2)$ , the instantaneous rate of change in the amount of dirt removed, in cubic meters per hour, at time  $t = 2$  hours.

## Part A

The point is earned for a correct expression that substitutes both correct values from the table, with or without simplification.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



|   |   |
|---|---|
| 0 | 1 |
|---|---|

The student response accurately includes a correct average rate of change.

## Solution:

$$\frac{f(12) - f(6)}{12 - 6} = \frac{g(12) - g(6)}{12 - 6} = \frac{474 - 306}{12 - 6} = \frac{168}{6} = 28$$

The average rate of change of the amount of dirt removed over the time interval  $6 \leq t \leq 12$  hours is 28 cubic meters per hour.

## 2-4-5

## Part B

The point is earned for a correct expression that substitutes both correct values from the table, with or without simplification.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



|   |   |
|---|---|
| 0 | 1 |
|---|---|

The student response accurately includes a correct approximation of  $f'(9)$ .

## Solution:

$$f'(9) \approx \frac{g(10) - g(8)}{10 - 8} = \frac{428 - 376}{10 - 8} = \frac{52}{2} = 26$$

## Part C

The response must explicitly connect differentiability and continuity to earn the first point and use the fact that  $f$  is a polynomial for  $0 \leq t < 6$  to explain why  $f$  is continuous on that interval to earn the second point. The third point is for continuity at  $t = 6$ .

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



|   |   |   |   |
|---|---|---|---|
| 0 | 1 | 2 | 3 |
|---|---|---|---|

The student response accurately includes all three of the criteria below.

- ☐  $f$  is continuous for  $0 \leq t < 6$
- ☐ differentiability of  $g$  implies continuity
- ☐  $\lim_{t \rightarrow 6^-} f(t) = \lim_{t \rightarrow 6^+} f(t) = f(6)$

## Solution:

Because  $f$  is a polynomial function for  $0 \leq t < 6$ ,  $f$  is continuous for  $0 \leq t < 6$ .

Because  $g$  is differentiable,  $g$  is continuous on  $6 \leq t \leq 12$ .  $f(t) = g(t)$  for  $6 \leq t \leq 12$ .

$$\lim_{t \rightarrow 6^-} f(t) = \lim_{t \rightarrow 6^-} \left( 48t + t^2 - \frac{t^3}{12} \right) = 48(6) + 6^2 - \frac{6^3}{12} = 288 + 36 - 18 = 306$$

$$\lim_{t \rightarrow 6^+} f(t) = f(6) = g(6) = 306$$

Because  $\lim_{t \rightarrow 6^-} f(t) = \lim_{t \rightarrow 6^+} f(t) = f(6)$ ,  $f$  is continuous at  $t = 6$ .

Thus,  $f$  is continuous for  $0 \leq t \leq 12$ .

## 2-4-5

## Part D

The first point is earned for demonstration of a correct derivative for  $f$  and does not require a separate statement of  $f'(t)$ , so  $f'(2) = 48 + 2 \cdot 2 - \frac{2^2}{4}$  earns both points. The derivative and evaluation of the derivative can be expressed in one line to earn both points.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



|   |   |   |
|---|---|---|
| 0 | 1 | 2 |
|---|---|---|

The student response accurately includes both of the criteria below.

☐  $f'(t)$

☐  $f'(2)$

**Solution:**

$$f(t) = 48t + t^2 - \frac{t^3}{12} \text{ for } 0 \leq t < 6$$

$$f'(t) = 48 + 2t - \frac{t^2}{4} \text{ for } 0 < t < 6$$

$$f'(2) = 48 + 2 \cdot 2 - \frac{2^2}{4} = 51$$

The instantaneous rate of change in the amount of dirt removed at  $t = 2$  hours is 51 cubic meters per hour.

## 2-4-5

## 62. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function  $f$  is assumed to be the set of all real numbers  $x$  for which  $f(x)$  is a real number.

$$f(t) = \begin{cases} g(t) & \text{for } 0 \leq t \leq 12 \\ \frac{t^2}{8} - 3t + 83 & \text{for } 12 < t \leq 24 \end{cases}$$

|                             |    |      |    |      |    |
|-----------------------------|----|------|----|------|----|
| $t$ (hours)                 | 0  | 4    | 6  | 8    | 12 |
| $g(t)$ (degrees Fahrenheit) | 87 | 81.5 | 76 | 70.5 | 65 |

The temperature of a room, in degrees Fahrenheit, on a certain day is modeled by the function  $f$  defined above, where  $g$  is a differentiable function and  $t$  is measured in hours. Values of  $g(t)$  at selected values of  $t$  are given in the table above.

- According to the model  $f$ , what is the average rate of change of the temperature of the room over the time interval  $0 \leq t \leq 12$  hours?
- Use the data in the table to approximate  $f'(10)$ , the instantaneous rate of change in the temperature of the room, in degrees Fahrenheit per hour, at time  $t = 10$  hours. Show the computations that lead to your answer.
- Is  $f$  continuous for  $0 \leq t \leq 24$ ? Justify your answer.
- Find  $f'(20)$ , the instantaneous rate of change in the temperature of the room, in degrees Fahrenheit per hour, at time  $t = 20$  hours.

## Part A

The point is earned for a correct expression that substitutes both correct values from the table, with or without simplification.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



|   |   |
|---|---|
| 0 | 1 |
|---|---|

The student response accurately includes a correct average rate of change.

## Solution:

$$\frac{f(12) - f(0)}{12 - 0} = \frac{g(12) - g(0)}{12 - 0} = \frac{65 - 87}{12} = -\frac{11}{6}$$

The average rate of change of the temperature of the room over the time interval  $0 \leq t \leq 12$  hours is  $-\frac{11}{6}$  degrees Fahrenheit per hour

## 2-4-5

## Part B

The second point is earned for a correct expression that substitutes both correct values from the table, with or without simplification.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



|   |   |
|---|---|
| 0 | 1 |
|---|---|

The student response accurately includes a correct approximation of  $f'(10)$

## Solution:

$$f'(10) \approx \frac{g(12) - g(8)}{12 - 8} = \frac{65 - 70.5}{4} = -\frac{5.5}{4}$$

## Part C

The response must explicitly connect differentiability and continuity to earn the first point and use the fact that  $f$  is a polynomial on  $12 < t \leq 24$  to explain why  $f$  is continuous on that interval to earn the second point. The third point is for continuity at  $t = 12$ .

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



|   |   |   |   |
|---|---|---|---|
| 0 | 1 | 2 | 3 |
|---|---|---|---|

The student response accurately includes all three of the criteria below.

- ☐ differentiability of  $g$  implies continuity
- ☐  $f$  is continuous on  $12 < t \leq 24$
- ☐  $\lim_{t \rightarrow 12^-} f(t) = \lim_{t \rightarrow 12^+} f(t) = f(12)$

## Solution:

Because  $g$  is differentiable,  $g$  is continuous on  $0 \leq t \leq 12$ .  $f(t) = g(t)$  for  $0 \leq t \leq 12$ .

Because  $f$  is a polynomial function on  $12 < t \leq 24$ ,  $f$  is continuous on  $12 < t \leq 24$ .

$$\lim_{t \rightarrow 12^-} f(t) = f(12) = g(12) = 65$$

$$\lim_{t \rightarrow 12^+} f(t) = \frac{12^2}{8} - 3(12) + 83 = 65$$

Because  $\lim_{t \rightarrow 12^-} f(t) = \lim_{t \rightarrow 12^+} f(t) = f(12)$ ,  $f$  is continuous at  $t = 12$ .

**2-4-5**

Thus,  $f$  is continuous on  $0 \leq t \leq 24$ .

**Part D**

The first point is earned for demonstration of a correct derivative for  $f$  and does not require a separate statement of  $f'(t)$ . The derivative and evaluation of the derivative can be expressed in one line to earn both points.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



|   |   |   |
|---|---|---|
| 0 | 1 | 2 |
|---|---|---|

The student response accurately includes both of the criteria below.

- ☐  $f'(t)$
- ☐  $f'(20)$

**Solution:**

$$f(t) = \frac{t^2}{8} - 3t + 83$$

$$f'(t) = \frac{1}{4}t - 3$$

$$f'(20) = \frac{1}{4}(20) - 3 = 5 - 3 = 2$$

The instantaneous rate of change in the temperature of the room at  $t = 20$  hours is 2 degrees Fahrenheit per hour.

## 2-4-5

## 63. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function  $f$  is assumed to be the set of all real numbers  $x$  for which  $f(x)$  is a real number.

|                      |   |    |     |     |     |     |
|----------------------|---|----|-----|-----|-----|-----|
| $t$ (minutes)        | 0 | 15 | 45  | 70  | 90  | 100 |
| $A(t)$ (automobiles) | 0 | 30 | 190 | 250 | 405 | 600 |

Prior to a sporting event, the number of automobiles that have entered a stadium parking lot is modeled by the differentiable function  $A$ , where  $t$  is the number of minutes since the parking lot opened. Values of  $A(t)$  for selected values of  $t$  are given in the table above.

According to the model, what is the average rate at which automobiles enter the parking lot, in automobiles per minute, over the time interval  $45 \leq t \leq 90$  minutes?

Write  $A'(95)$  as the limit of a difference quotient. Use the data in the table to approximate  $A'(95)$ . Show the computations that lead to your answer.

What is the shortest time interval during which it is guaranteed that  $A(t) = 400$  for some time  $t$  in the interval? Justify your answer.

For  $0 \leq t \leq 45$ , the function  $f$  defined by  $f(t) = \frac{2}{27}t^2 + \frac{8}{9}t$  models the number of automobiles that have entered the parking lot, where  $t$  is the number of minutes since the parking lot opened. Find  $f'(10)$ , the rate at which automobiles enter the parking lot, in automobiles per minute, at time  $t = 10$  minutes.

**Part A**

The point is earned for a correct expression that substitutes both correct values from the table, with or without simplification.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



|   |   |
|---|---|
| 0 | 1 |
|---|---|

The student response accurately includes a correct average rate of change.

**Solution:**

$$\frac{A(90) - A(45)}{90 - 45} = \frac{405 - 190}{90 - 45} = \frac{215}{45} = \frac{43}{9}$$

The average rate at which automobiles enter the parking the time interval  $45 \leq t \leq 90$  minutes is  $\frac{43}{9}$  automobiles per minute.

## 2-4-5

## Part B

The second point is earned for a correct expression that substitutes both correct values from the table, with or without simplification.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



|   |   |   |
|---|---|---|
| 0 | 1 | 2 |
|---|---|---|

The student response accurately includes both of the criteria below.

- ☐ limit expression
- ☐ approximation

## Solution:

$$A'(95) = \lim_{t \rightarrow 95} \frac{A(t) - A(95)}{t - 95}$$

$$\text{OR } A'(95) = \lim_{h \rightarrow 0} \frac{A(95+h) - A(95)}{h}$$

$$A'(95) \approx \frac{A(100) - A(90)}{100 - 90} = \frac{600 - 405}{100 - 90} = \frac{195}{10} = \frac{39}{2}$$

## Part C

1 out of 2 points if correct justification does not specifically reference IVT.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



|   |   |   |
|---|---|---|
| 0 | 1 | 2 |
|---|---|---|

The student response accurately includes both of the criteria below.

- ☐ time interval
- ☐ justification, using Intermediate Value Theorem

## Solution:

The shortest time interval for which it is guaranteed that  $A(t) = 400$  is  $70 \leq t \leq 90$ .

Because  $A$  is differentiable,  $A$  is continuous.

$$A(70) = 250 < 400 < 405 = A(90)$$

## 2-4-5

By the Intermediate Value Theorem, there must be at least one time  $t$ , for  $70 \leq t \leq 90$ , such that  $A(t) = 400$ .

**Part D**

The first point is earned for demonstration of a correct derivative for  $f$  and does not require a separate statement of  $f'(t)$ . The derivative and evaluation of the derivative can be expressed in one line to earn both points.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



|   |   |   |
|---|---|---|
| 0 | 1 | 2 |
|---|---|---|

The student response accurately includes both of the criteria below.

- ☐  $f'(t)$
- ☐  $f'(10)$

**Solution:**

$$f'(t) = \frac{4}{27}t + \frac{8}{9}$$

$$f'(10) = \frac{4}{27} \cdot 10 + \frac{8}{9} = \frac{40}{27} + \frac{24}{27} = \frac{64}{27}$$

At time  $t = 10$  minutes, automobiles enter the parking lot at a rate of  $\frac{64}{27}$  automobiles per minute.

## 2-4-5

## 64. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function  $f$  is assumed to be the set of all real numbers  $x$  for which  $f(x)$  is a real number.

|                     |   |    |     |     |     |
|---------------------|---|----|-----|-----|-----|
| $t$ (hours)         | 0 | 1  | 3   | 6   | 8   |
| $P(t)$ (passengers) | 0 | 35 | 204 | 600 | 728 |

The number of passengers who have boarded a ship is modeled by the differentiable function  $P$ , where  $t$  is the number of hours since boarding began. Values of  $P(t)$  for selected values of  $t$  are given in the table above.

- According to the model, what is the average rate at which passengers board the ship, in passengers per hour, over the time interval  $1 \leq t \leq 8$  hours?
- Write  $P'(4.5)$  as the limit of a difference quotient. Use the data in the table to approximate  $P'(4.5)$ . Show the computations that lead to your answer.
- Must there be a time  $t$ , for  $3 \leq t \leq 6$ , at which  $P(t) = 500$ ? Justify your answer.
- The total number of gallons of water used by the passengers on the ship is modeled by the function  $G$  defined by  $G(t) = 120t\sqrt{t}$  for  $0 \leq t \leq 8$ , where  $t$  is the number of hours since boarding began. Find  $G'(4)$ , the rate at which passengers use water, in gallons per hour, at time  $t = 4$  hours.

## Part A

The point is earned for a correct expression that substitutes both correct values from the table, with or without simplification.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

|   |   |   |
|---|---|---|
|   |   | ✓ |
| 0 | 1 |   |

The student response accurately includes a correct average rate of change.

## Solution:

$$\frac{P(8) - P(1)}{8 - 1} = \frac{728 - 35}{7} = 99$$

The average rate at which passengers board the ship over the time interval  $1 \leq t \leq 8$  hours is 99 passengers per hour.

## Part B

## 2-4-5

The second point is earned for a correct expression that substitutes both correct values from the table, with or without simplification.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



|   |   |   |
|---|---|---|
| 0 | 1 | 2 |
|---|---|---|

The student response accurately includes both of the criteria below.

- ☐ limit expression
- ☐ approximation

**Solution:**

$$P'(4.5) = \lim_{t \rightarrow 4.5} \frac{P(t) - P(4.5)}{t - 4.5}$$

OR

$$P'(4.5) = \lim_{h \rightarrow 0} \frac{P(4.5+h) - P(4.5)}{h}$$

$$P'(4.5) \approx \frac{P(6) - P(3)}{6 - 3} = \frac{600 - 204}{3} = 132$$

**Part C**

1 out of 2 points if correct justification does not specifically reference IVT.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response



|   |   |   |
|---|---|---|
| 0 | 1 | 2 |
|---|---|---|

The student response accurately includes both of the criteria below.

- ☐  $P$  is continuous with reason
- ☐ justification, using Intermediate Value Theorem

**Solution:**

Because  $P$  is differentiable,  $P$  is continuous.

$$P(3) = 204 < 500 < 600 = P(6)$$

By the Intermediate Value Theorem, there must be at least one time  $t$ , for  $3 \leq t \leq 6$ , such that

$$P(t) = 500.$$

## 2-4-5

## Part D

The first point is earned for demonstration of a correct derivative for  $G$  and does not require a separate statement of  $G'(t)$ .

The derivative and evaluation of the derivative can be expressed in one line to earn both points.



|   |   |   |
|---|---|---|
| 0 | 1 | 2 |
|---|---|---|

The student response accurately includes both of the criteria below.

☐  $G'(t)$

☐  $G'(4)$

**Solution:**

$$G(t) = 120t\sqrt{t} = 120t^{\frac{3}{2}}$$

$$G'(t) = 120 \cdot \frac{3}{2}t^{\frac{1}{2}} = 180\sqrt{t}$$

$$G'(4) = 180\sqrt{4} = 180 \cdot 2 = 360$$

At time  $t = 4$  hours, passengers use water at a rate of 360 gallons per hour.

## 2-4-5

65. Let  $f$  be the function defined as shown.

$$f(x) = \begin{cases} 1 & \text{if } x < 0 \\ x^2 + 1 & \text{if } 0 \leq x \leq 1 \\ x + 1 & \text{if } x > 1 \end{cases}$$

Which of the following statements is true?

- (A)  $f$  is continuous and differentiable for all  $x$ .  
 (B)  $f$  is continuous for all  $x$  and differentiable except at  $x = 1$ . ✓  
 (C)  $f$  is continuous for all  $x$  and differentiable except at  $x = 0$  and  $x = 1$ .  
 (D)  $f$  is continuous except at  $x = 1$ .

### Answer B

Correct. The piecewise-defined function  $f$  is continuous at  $x = 0$  because the functions  $1$  and  $x^2 + 1$  are equal at the boundary  $x = 0$ , and the common value is  $f(0) = 1$ . The piecewise-defined function  $f$  is continuous at  $x = 1$  because the functions  $x^2 + 1$  and  $x + 1$  are equal at the boundary  $x = 1$ , and the common value is  $f(1) = 2$ . The function  $f$  is continuous at all other values of  $x$  because each individual piece is continuous at all other values of  $x$ . Hence  $f$  is continuous for all  $x$ .

The function  $f$  is differentiable at  $x = 0$  since  $\lim_{h \rightarrow 0^+} \frac{f(h) - f(0)}{h} = \lim_{h \rightarrow 0^+} \frac{h^2 + 1 - 1}{h} = \lim_{h \rightarrow 0^+} h = 0$  and  $\lim_{h \rightarrow 0^-} \frac{f(h) - f(0)}{h} = \lim_{h \rightarrow 0^-} \frac{1 - 1}{h} = 0$ . Therefore  $f'(0) = 0$ .

The function  $f$  is not differentiable at  $x = 1$  since  $\lim_{h \rightarrow 0^+} \frac{f(1+h) - f(1)}{h} = \lim_{h \rightarrow 0^+} \frac{(1+h+1) - 2}{h} = \lim_{h \rightarrow 0^+} 1 = 1$  but  $\lim_{h \rightarrow 0^-} \frac{f(1+h) - f(1)}{h} = \lim_{h \rightarrow 0^-} \frac{(1+h)^2 + 1 - 2}{h} = \lim_{h \rightarrow 0^-} \frac{h^2 + 2h}{h} = \lim_{h \rightarrow 0^-} (h + 2) = 2$ .

Since the left- and right-sided limits are not equal, the limit of the difference quotient does not exist at  $x = 1$ .

The function  $f$  is differentiable at all other values of  $x$  because each individual piece is differentiable at all other values of  $x$ .

An alternative way to see that the piecewise-defined function is differentiable at  $x = 0$  but not at  $x = 1$  is to observe that

$$f'(x) = \begin{cases} 0 & \text{if } x < 0 \\ 2x & \text{if } 0 < x < 1. \\ 1 & \text{if } x > 1 \end{cases}$$

Since  $f$  is continuous at  $x = 0$  and the derivatives  $0$  and  $2x$  are equal at  $x = 0$ ,  $f$  is differentiable at  $x = 0$ . But since  $2x$  and  $1$  are not equal at  $x = 1$ ,  $f$  is not differentiable at  $x = 1$ .

For details on the alternative method, see the article “Differentiability of Piecewise Defined Functions” in the Classroom Resources section of the Calculus AB or Calculus BC course page at AP Central.

## 2-4-5

66. What is the value of  $\lim_{h \rightarrow 0} \frac{(16+h)^{\frac{1}{4}} - 2}{h}$ ?

(A) 0

(B)  $\frac{1}{32}$

(C)  $\frac{1}{8}$

(D) 1

**Answer B**

Correct. This is the definition of the derivative of the function  $f(x) = x^{\frac{1}{4}}$  at  $x = 16$ , denoted by  $f'(16)$ . Application of the power rule gives  $f'(x) = \frac{1}{4}x^{-\frac{3}{4}} = \frac{1}{4x^{\frac{3}{4}}}$ . Therefore,  $f'(16) = \frac{1}{4(16)^{\frac{3}{4}}} = \frac{1}{4(8)} = \frac{1}{32}$ .

67. 
$$g(x) = \begin{cases} 3 - x & \text{for } x < 3 \\ 4x - 12 & \text{for } x \geq 3 \end{cases}$$

If  $g$  is the function defined above, then  $g'(3)$  is

(A)  $-1$

(B) 0

(C) 4

(D) nonexistent

**Answer D**

Correct. The graph of  $g$  consists of two rays that meet at point  $(3, 0)$ . The graph has a sharp corner at  $x = 3$ , and thus the derivative of  $g$  does not exist at  $x = 3$ . More formally, the derivative of  $g$  at  $x = 3$  does not exist because

$\lim_{h \rightarrow 0^-} \frac{g(3+h) - g(3)}{h} = -1 \neq 4 = \lim_{h \rightarrow 0^+} \frac{g(3+h) - g(3)}{h}$ , and therefore  $\lim_{h \rightarrow 0} \frac{g(3+h) - g(3)}{h}$  does not exist.

Alternatively, for  $x < 3$ , it follows from the power rule that  $g'(x) = -1$ . And for  $x > 3$ , it follows from the power rule that  $g'(x) = 4$ . Therefore  $\lim_{x \rightarrow 3^-} g'(x) \neq \lim_{x \rightarrow 3^+} g'(x)$ , and thus  $g'(3)$  does not exist.

## 2-4-5

68. What is the value of  $\lim_{h \rightarrow 0} \frac{(8+h)^{\frac{1}{3}} - 2}{h}$ ?

(A) 0

(B)  $\frac{1}{12}$

(C)  $\frac{1}{4}$

(D) 1

**Answer B**

Correct. This is the definition of the derivative of the function  $f(x) = x^{\frac{1}{3}}$  at  $x = 8$ , denoted by  $f'(8)$ . Application of the power rule gives  $f'(x) = \frac{1}{3}x^{-\frac{2}{3}} = \frac{1}{3x^{\frac{2}{3}}}$ . Therefore,  $f'(8) = \frac{1}{3 \cdot 8^{\frac{2}{3}}} = \frac{1}{3 \cdot 4} = \frac{1}{12}$ .

69.  $\lim_{h \rightarrow 0} \frac{(2+h)^3 + 2(2+h) - (2)^3 - 2(2)}{h}$  is

(A) 6

(B) 12

(C) 14

(D) nonexistent

**Answer C**

Correct. The limit is the definition of the derivative of the function  $f(x) = x^3 + 2x$  at  $x = 2$ , denoted by  $f'(2)$ . Application of the power rule gives  $f'(x) = 3x^2 + 2$ . Therefore,  $f'(2) = 3(2)^2 + 2 = 14$ .

The limit could also be evaluated by using algebraic manipulation, as follows.

$$\lim_{h \rightarrow 0} \frac{(2+h)^3 + 2(2+h) - (2)^3 - 2(2)}{h} = \lim_{h \rightarrow 0} \frac{8+12h+6h^2+h^3+4+2h-8-4}{h} = \lim_{h \rightarrow 0} \frac{14h+6h^2+h^3}{h} = \lim_{h \rightarrow 0} (14 + 6h + h^2) = 14 + 0 + 0 = 14$$

2-4-5

70.

$$g(x) = \begin{cases} 2 - 2x & \text{for } x < 1 \\ 5x - 5 & \text{for } x \geq 1 \end{cases}$$

If  $g$  is the function defined above, then  $g'(1)$  is

- (A)  $-2$   
 (B)  $0$   
 (C)  $5$   
 (D) nonexistent



### Answer D

Correct. The graph of  $g$  consists of two rays that meet at point  $(1, 0)$ . The graph has a sharp corner at  $x = 1$ , and thus the derivative of  $g$  does not exist at  $x = 1$ . More formally, the derivative of  $g$  at  $x = 1$  does not exist because

$$\lim_{h \rightarrow 0^-} \frac{g(1+h) - g(1)}{h} = -2 \neq 5 = \lim_{h \rightarrow 0^+} \frac{g(1+h) - g(1)}{h},$$

and therefore,  $\lim_{h \rightarrow 0} \frac{g(1+h) - g(1)}{h}$  does not exist. Alternatively, for  $x < 1$ , it follows from the power rule that  $g'(x) = -2$ . And

for  $x > 1$ , it follows from the power rule that  $g'(x) = 5$ . Therefore,

$$\lim_{x \rightarrow 1^-} g'(x) \neq \lim_{x \rightarrow 1^+} g'(x),$$

and thus  $g'(1)$  does not exist.

71.

$$\lim_{h \rightarrow 0} \frac{(x+h)^2 + 4(x+h) - x^2 - 4x}{h} \text{ is}$$

- (A)  $x^3 + x^4$   
 (B)  $x^2 + 4x$   
 (C)  $2x + 4$   
 (D) nonexistent



### Answer C

Correct. The limit is the definition of the derivative of the function  $f(x) = x^2 + 4x$ . Application of the power rule gives  $f'(x) = 2x + 4$ .

The limit could also be evaluated by using algebraic manipulation, as follows.

$$\lim_{h \rightarrow 0} \frac{(x+h)^2 + 4(x+h) - x^2 - 4x}{h} = \lim_{h \rightarrow 0} \frac{x^2 + 2xh + h^2 + 4x + 4h - x^2 - 4x}{h} = \lim_{h \rightarrow 0} \frac{2xh + h^2 + 4h}{h} = \lim_{h \rightarrow 0} (2x + h + 4) = 2x + 4$$

## 2-4-5

72. If  $f(x) = \frac{1}{x^6}$ , then  $f'(x) =$

- (A)  $\frac{1}{6x^5}$   
(B)  $-\frac{6}{x^5}$   
(C)  $-\frac{1}{6x^7}$   
(D)  $-\frac{6}{x^7}$

**Answer D**

Correct. Writing  $\frac{1}{x^6}$  as  $x^{-6}$ , the power rule for differentiation gives  $f'(x) = -6x^{-7} = -\frac{6}{x^7}$ .

73. If  $f(x) = x^5$ , then  $f'(x) =$

- (A)  $x^4$   
(B)  $4x^4$   
(C)  $5x^4$   
(D)  $5x^5$

**Answer C**

Correct. According to the power rule for differentiation, the derivative of functions of the form  $f(x) = x^r$  is  $f'(x) = rx^{r-1}$ .

74. If  $f(x) = \frac{1}{x^7}$ , then  $f'(x) =$

- (A)  $\frac{1}{7x^6}$   
(B)  $-\frac{7}{x^6}$   
(C)  $-\frac{1}{7x^8}$   
(D)  $-\frac{7}{x^8}$

**Answer D**

Correct. Writing  $\frac{1}{x^7}$  as  $x^{-7}$ , the power rule for differentiation gives  $f'(x) = -7x^{-8} = -\frac{7}{x^8}$ .

## 2-4-5

75. If  $f$  is the function defined by  $f(x) = \sqrt[4]{x}$ , what is  $f'(x)$ ?

- (A)  $\frac{1}{4}x^{\frac{1}{4}}$   
(B)  $x^{-\frac{3}{4}}$   
(C)  $\frac{1}{4}x^{-\frac{3}{4}}$  ✓  
(D)  $4 \cdot \sqrt[3]{x}$

**Answer C**

Correct.  $f(x) = \sqrt[4]{x} = x^{\frac{1}{4}}$  is a power function. Therefore, the power rule is applied to determine the derivative.

$$f'(x) = \frac{d}{dx} \left( x^{\frac{1}{4}} \right) = \frac{1}{4} x^{-\frac{3}{4}}$$

76. If  $f(x) = x^6$ , then  $f'(x) =$

- (A)  $x^5$   
(B)  $5x^5$   
(C)  $6x^5$  ✓  
(D)  $6x^6$

**Answer C**

Correct. According to the power rule for differentiation, the derivative of functions of the form  $f(x) = x^r$  is  $f'(x) = rx^{r-1}$ .

77. If  $f$  is the function defined by  $f(x) = \sqrt[3]{x}$ , what is  $f'(x)$ ?

- (A)  $\frac{1}{3}x^{\frac{1}{3}}$   
(B)  $x^{-\frac{2}{3}}$   
(C)  $\frac{1}{3}x^{-\frac{2}{3}}$  ✓  
(D)  $3 \cdot \sqrt{x}$

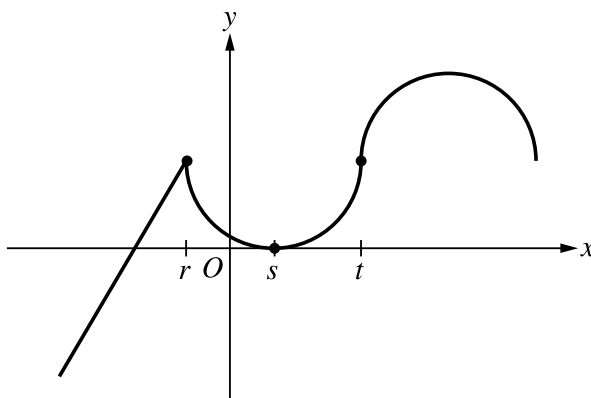
**Answer C**

Correct.  $f(x) = \sqrt[3]{x} = x^{\frac{1}{3}}$  is a power function. Therefore, the power rule is applied to determine the derivative.

$$f'(x) = \frac{d}{dx} \left( x^{\frac{1}{3}} \right) = \frac{1}{3} x^{-\frac{2}{3}}$$

2-4-5

78.

Graph of  $f$ 

The figure shows the graph of a continuous piecewise-defined function  $f$ . The graph consists of a line segment and two semicircles. For which of the following values of  $x$  is  $f'(x)$  undefined?

- (A)  $r$  only
- (B)  $t$  only
- (C)  $r$  and  $s$
- (D)  $r$  and  $t$



### Answer D

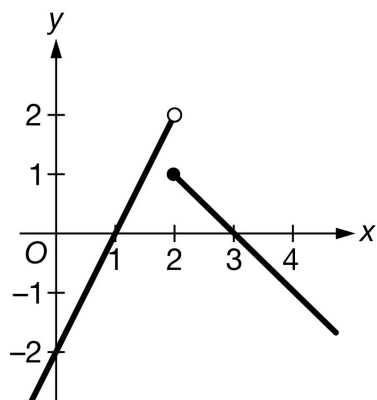
Correct.  $\lim_{h \rightarrow 0^+} \frac{f(r+h)-f(r)}{h}$  does not exist, since the difference quotient approaches  $-\infty$  as  $h$  approaches 0 from the right and the secant lines become more and more vertical as the point at  $x = r + h$  approaches the endpoint of the semicircle. Therefore,  $\lim_{h \rightarrow 0} \frac{f(r+h)-f(r)}{h}$  does not exist and so  $f'(x)$  is undefined at  $x = r$ . Visually, the fact that  $f'(x)$  is undefined at  $x = r$  is suggested by the sharp turn as the graph changes from the line to the semicircle at  $x = r$ .

Neither  $\lim_{h \rightarrow 0^-} \frac{f(t+h)-f(t)}{h}$  nor  $\lim_{h \rightarrow 0^+} \frac{f(t+h)-f(t)}{h}$  exist, since in both cases the difference quotient approaches  $+\infty$  as the secant lines become more and more vertical. Therefore,  $\lim_{h \rightarrow 0} \frac{f(t+h)-f(t)}{h}$  does not exist and so  $f'(x)$  is also undefined at  $x = t$ .

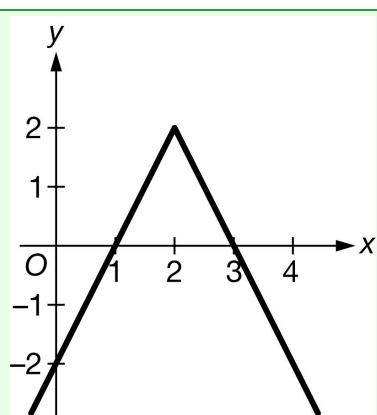
## 2-4-5

79. Let  $g$  be a function such that  $\lim_{x \rightarrow 2^-} g(x) = g(2)$  and  $\lim_{x \rightarrow 2^+} g'(x) \neq \lim_{x \rightarrow 2^-} g'(x)$ . Which of the following could be the graph of  $g$ ?

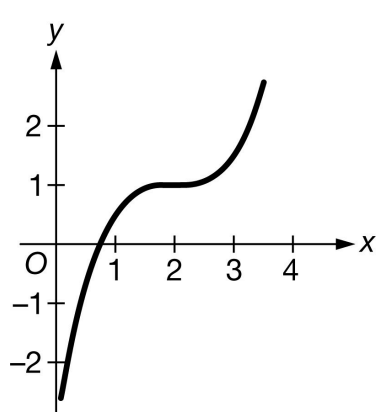
(A)



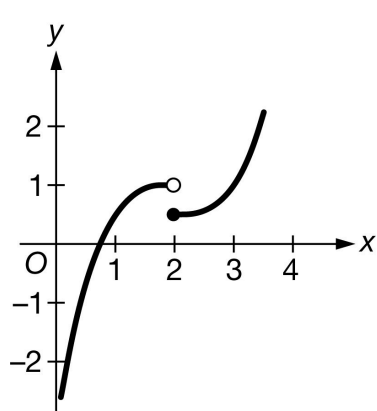
(B)



(C)



(D)



2-4-5

**Answer B**

Correct. Since  $\lim_{x \rightarrow 2} g(x) = g(2)$ , the function  $g$  is continuous at  $x = 2$ . However, since  $\lim_{x \rightarrow 2^+} g'(x) \neq \lim_{x \rightarrow 2^-} g'(x)$ , the function  $g$  is not differentiable at  $x = 2$ . This graph is continuous at  $x = 2$ , but it is not differentiable at  $x = 2$  because the slopes of the two line segments on either side of  $x = 2$  are not equal. Therefore, this could be the graph of  $g$ .

80. A particle moves along the  $y$ -axis so that at time  $t \geq 0$  its position is given by  $y(t) = t^3 - 4t^2 + 4t + 3$ . Which of the following statements describes the motion of the particle at time  $t = 1$ ?

(A) The particle is moving down the  $y$ -axis with decreasing velocity. ✓

(B) The particle is moving down the  $y$ -axis with increasing velocity.

(C) The particle is moving up the  $y$ -axis with decreasing velocity.

(D) The particle is moving up the  $y$ -axis with increasing velocity.

**Answer A**

Correct. The velocity of the particle is  $v(t) = 3t^2 - 8t + 4$ . At time  $t = 1$ ,  $v(1) = -1$ . Since the velocity is negative, the particle is moving down the  $y$ -axis. The rate of change of the velocity is  $v'(t) = 6t - 8$ . At time  $t = 1$ ,  $v'(1) = -2$ . Since this is negative, the particle is moving with decreasing velocity at time  $t = 1$ .

## 2-4-5

81. Let  $f$  be the function given by  $f(x) = \sqrt{x^4 - 16x^2}$ .
- Find the domain of  $f$ .
  - Describe the symmetry, if any, of the graph of  $f$ .
  - Find  $f'(x)$ .
  - Find the slope of the line normal to the graph of  $f$  at  $x = 5$ .

## Part A

1 point is earned for radicand  $\geq 0$

1 point is earned for  $|x| \geq 4$

1 point is earned for  $x = 0$

$$x^4 - 16x^2 \geq 0$$

$$x^2(x^2 - 16) \geq 0$$

$$x^2 \geq 16 \text{ or } x = 0$$

$$|x| \geq 4 \text{ or } x = 0$$



|   |   |   |   |
|---|---|---|---|
| 0 | 1 | 2 | 3 |
|---|---|---|---|

The student response earns all of the following points:

1 point is earned for radicand  $\geq 0$

1 point is earned for  $|x| \geq 4$

1 point is earned for  $x = 0$

$$x^4 - 16x^2 \geq 0$$

$$x^2(x^2 - 16) \geq 0$$

$$x^2 \geq 16 \text{ or } x = 0$$

$$|x| \geq 4 \text{ or } x = 0$$

## Part B

1 point is earned for the correct answer

Symmetric about the  $y$ -axis



|   |   |
|---|---|
| 0 | 1 |
|---|---|

**2-4-5**

The student response earns all of the following points:

1 point is earned for the correct answer

Symmetric about the  $y$ -axis

**Part C**

3 points are earned for correct derivative

$$\begin{aligned} f'(x) &= \frac{1}{2}(x^4 - 16x^2)^{-\frac{1}{2}}(4x^3 - 32x) \\ &= \frac{2x(x^2 - 8)}{|x|\sqrt{x^2 - 16}} \end{aligned}$$

✓

|   |   |   |   |
|---|---|---|---|
| 0 | 1 | 2 | 3 |
|---|---|---|---|

The student response earns all of the following points:

3 points are earned for correct derivative

$$\begin{aligned} f'(x) &= \frac{1}{2}(x^4 - 16x^2)^{-\frac{1}{2}}(4x^3 - 32x) \\ &= \frac{2x(x^2 - 8)}{|x|\sqrt{x^2 - 16}} \end{aligned}$$

**Part D**

1 point is earned for evaluating  $f'(x)$  found in (c) at  $x = 5$

1 point is earned for slope of normal

$$\begin{aligned} f'(5) &= \frac{2(5)(25-8)}{5\sqrt{25-16}} \\ &= \frac{10(17)}{5\sqrt{9}} = \frac{170}{15} = \frac{34}{3} \end{aligned}$$

$$\therefore \text{slope of normal line is } -\frac{3}{34}$$

✓

|   |   |   |
|---|---|---|
| 0 | 1 | 2 |
|---|---|---|

**2-4-5**

The student response earns all of the following points:

1 point is earned for evaluating  $f'(x)$  found in (c) at  $x = 5$

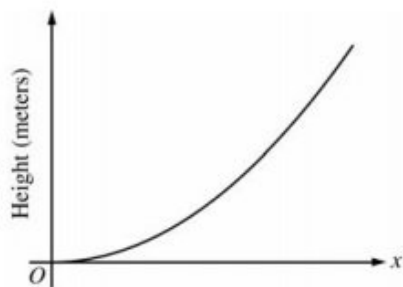
1 point is earned for slope of normal

$$\begin{aligned} f'(5) &= \frac{2(5)(25-8)}{5\sqrt{25-16}} \\ &= \frac{10(17)}{5\sqrt{9}} = \frac{170}{15} = \frac{34}{3} \end{aligned}$$

$$\therefore \text{slope of normal line is } -\frac{3}{34}$$

2-4-5

82.



The figure above is the graph of a function of  $x$ , which models the height of a skateboard ramp. The function meets the following requirements.

- (i) At  $x = 0$ , the value of the function is 0, and the slope of the graph of the function is 0.
  - (ii) At  $x = 4$ , the value of the function is 1, and the slope of the graph of the function is 1.
  - (iii) Between  $x = 0$  and  $x = 4$ , the function is increasing.
- (a) Let  $f(x) = ax^2$ , where  $a$  is a nonzero constant. Show that it is not possible to find a value for  $a$  so that  $f$  meets requirement (ii) above.
- (b) Let  $g(x) = cx^3 - \frac{x^2}{16}$ , where  $c$  is a nonzero constant. Find the value of  $c$  so that  $g$  meets requirement (ii) above. Show the work that leads to your answer.
- (c) Using the function  $g$  and your value of  $c$  from part (b), show that  $g$  does not meet requirement (iii) above.
- (d) Let  $h(x) = x^n/k$ , where  $k$  is a nonzero constant and  $n$  is a positive integer. Find the values of  $k$  and  $n$  so that  $h$  meets requirement (ii) above. Show that  $h$  also meets requirements (i) and (iii) above.

**Part A**

1 point is earned for if  $a = 1/16$  or  $a = 1/8$

1 point is earned if shows  $a$  does not work

$f(4) = 1$  implies that  $a = 1/16$  and  $f'(4) = 2a(4) = 1$  implies that  $a = 1/8$ . Thus,  $f$  cannot satisfy (ii).



|   |   |   |
|---|---|---|
| 0 | 1 | 2 |
|---|---|---|

The student response earns all of the following points:

1 point is earned for if  $a = 1/16$  or  $a = 1/8$

1 point is earned if shows  $a$  does not work

$f(4) = 1$  implies that  $a = 1/16$  and  $f'(4) = 2a(4) = 1$  implies that  $a = 1/8$ . Thus,  $f$  cannot satisfy (ii).

**Part B**

1 point is earned for the value of  $c$

## 2-4-5

$$g(4) = 64c - 1 = 1 \text{ implies that } c = \frac{1}{32}.$$

$$\text{When } c = \frac{1}{32}, g'(4) = 3c(4)^2 - \frac{2(4)}{16} = 3\left(\frac{1}{32}\right)(16) - \frac{1}{2} = 1$$



|   |   |
|---|---|
| 0 | 1 |
|---|---|

The student response earns all of the following points:

1 point is earned for the value of  $c$

$$g(4) = 64c - 1 = 1 \text{ implies that } c = \frac{1}{32}.$$

$$\text{When } c = \frac{1}{32}, g'(4) = 3c(4)^2 - \frac{2(4)}{16} = 3\left(\frac{1}{32}\right)(16) - \frac{1}{2} = 1$$

**Part C**

1 point is earned for  $g'(x)$

1 point is earned for the explanation

$$g'(x) = \frac{3}{32}x^2 - \frac{x}{8} = \frac{1}{32}x(3x - 4)$$

$g'(x) < 0$  for  $0 < x < 4/3$ , so  $g$  does not satisfy (iii).



|   |   |   |
|---|---|---|
| 0 | 1 | 2 |
|---|---|---|

The student response earns all of the following points:

1 point is earned for  $g'(x)$

1 point is earned for the explanation

$$g'(x) = \frac{3}{32}x^2 - \frac{x}{8} = \frac{1}{32}x(3x - 4)$$

**2-4-5**

$g'(x) < 0$  for  $0 < x < 4/3$ , so  $g$  does not satisfy (iii).

**Part D**

1 point is earned for  $\frac{4^n}{k} = 1$

1 point is earned for  $\frac{n4^{n-1}}{k} = 1$

1 point is earned for values for  $k$  and  $n$

1 point is earned for verification

$$h(4) = \frac{4^n}{k} = 1 \text{ implies that } 4^n = k.$$

$$h'(4) = \frac{n4^{n-1}}{k} = \frac{n4^{n-1}}{4^n} = \frac{n}{4} = 1 \text{ gives } n = 4 \text{ and } k = 4^4 = 256.$$

$$h(x) = \frac{x^4}{256} \Rightarrow h(0) = 0.$$

$$h'(x) = \frac{4x^3}{256} \Rightarrow h'(0) = 0 \text{ and } h'(x) > 0 \text{ for } 0 < x < 4.$$



|   |   |   |   |   |
|---|---|---|---|---|
| 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|

The student response earns all of the following points:

1 point is earned for  $\frac{4^n}{k} = 1$

1 point is earned for  $\frac{n4^{n-1}}{k} = 1$

1 point is earned for values for  $k$  and  $n$

1 point is earned for verification

$$h(4) = \frac{4^n}{k} = 1 \text{ implies that } 4^n = k.$$

$$h'(4) = \frac{n4^{n-1}}{k} = \frac{n4^{n-1}}{4^n} = \frac{n}{4} = 1 \text{ gives } n = 4 \text{ and } k = 4^4 = 256.$$

**2-4-5**

$$h(x) = \frac{x^4}{256} \Rightarrow h(0) = 0.$$

$$h'(x) = \frac{4x^3}{256} \Rightarrow h'(0) = 0 \text{ and } h'(x) > 0 \text{ for } 0 < x < 4.$$